

Fines and Financial Wellbeing*

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November 7, 2024

Abstract

While survey evidence suggests widespread financial fragility in the U.S., causal evidence on the implications of typical, negative income shocks is scarce. I estimate the impact of speeding fines on household finances using administrative traffic citation records and a panel of credit reports. Event studies reveal that fines averaging \$195 are associated with a \$34 increase in unpaid bills in collections. Given additional evidence that fine payment explains this effect and that default is the “last resort” for households, I interpret this finding as suggesting rates of inability to meet unplanned expenses which are consistent with the survey evidence. I also find that fines are associated with longer-run declines in credit scores, borrowing limits, and the likelihood of appearing as employed in payroll records covering a subset of large, high-paying employers. This impact on employment situations appears attributable to the diminished financial position of households rather than, e.g., downstream license suspensions.

JEL Codes: G51, I32, K42

*I am grateful to Will Dobbie, Ilyana Kuziemko, David Lee, and Alex Mas for unrelenting advice and encouragement on this project. Mark Aguiar, David Arnold, Leah Boustan, Jessica Brown, Elizabeth Cascio, Felipe Goncalves, Elisa Jacome, Henrik Kleven, Erzo Luttmer, Atif Mian, Jonathan Morduch, Jack Mountjoy, Chris Neilson, Scott Nelson, Whitney Rosenbaum, Bruce Sacerdote, Jussipekka Salo, Doug Staiger, Owen Zidar, Jonathan Zinman, Nathan Zorzi, and seminar participants at Princeton, Georgetown McCourt, Rochester, Chicago Booth, BU, Dartmouth, NYU Furman, CEP, Vassar, and Opportunity Insights provided helpful comments. I thank Beth Allman for providing the citations data and important institutional information and numerous credit bureau employees for assistance with accessing the credit report data. I benefitted from generous financial support from Princeton University and Dartmouth College. Any errors are my own.

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1 Introduction

The ability of households to cope with adverse shocks has important implications for taxation and social insurance policies (e.g., [Baily 1978](#); [Chetty 2006](#)). Despite the prediction of canonical models that liquidity-constrained households anticipate income volatility by accumulating buffer stock savings ([Deaton 1991](#), [Carroll et al. 1992](#); [Carroll 1997](#)), recent evidence has highlighted the lack of precautionary savings in the United States ([Beshears et al., 2018](#)). Half of all households accumulated no savings in 2010 ([Lusardi, 2011](#)) and forty percent of Americans indicated an inability to cover an emergency \$400 expense using liquid savings in a 2017 survey ([FRBG, 2018](#)).

This survey result in particular has received significant attention from journalists and policymakers. While some have cited the survey as another symbol of growing inequality or as motivation for an expanded social safety net, others have questioned the credibility of this statistic. Criticism has focused on the potentially misleading presentation of survey results ([Reynolds, 2019](#)), conflicting evidence from other data sources (e.g., [Chen 2019](#); [Nova 2019](#); [Bhutta & Dettling 2018](#)), and the belief that resilience against real world shocks may differ from self-reported ability to pay on a low-stakes survey ([Strain, 2019](#)).

An important obstacle in this debate is the lack of causal evidence on the impacts of typical, negative shocks on households. While ethnographies provide compelling accounts of families derailed by unplanned expenses (e.g., [Shipler 2004](#); [Desmond 2016](#)), the lack of credible variation in small income shocks and data on the household finances of lower-income populations have proven important obstacles to estimating causal effects. Existing studies have examined consumption responses to small positive shocks such as tax refunds (e.g., [Parker 2017](#)) or significant negative shocks such as hospital admissions ([Dobkin et al. 2018](#)) or job loss ([Stephens 2001](#), [Keys 2017](#)). Moreover, the literature’s reliance on policy variation generated by tax rebates or mortgage programs and on data from credit cards and bankruptcy filings has left the bottom end of the income distribution relatively understudied.

In this paper, I explore whether households are able to absorb unplanned shocks by estimating the impacts of fines for traffic violations on household financial situations. This setting has several important advantages. First, traffic fines represent a common form of everyday, unplanned expense that can be observed and measured in data. Over forty million citations are issued annually for speed limit violations alone and standard fines are well within the range of typical monthly income fluctuations ([Morduch & Schneider, 2016](#)). Second, as shown in [figure 1](#), policing activity disproportionately affects poorer communities, allowing for the study of a large sample of low-income households. Third, driver license suspensions imposed for nonpayment incentivize high payment rates on average, increasing confidence that a traffic ticket represents a true expense, or transitory income, shock.

To estimate the impact of fines, I link administrative data on the universe of traffic citations issued in Florida over 2011-2015 to a quarterly panel of credit reports for cited

drivers. The citations data provide near-complete coverage of the state’s traffic offenders and my analysis sample represents about three percent of Florida’s driving-age population. Credit reports offer a detailed account of an individual’s financial situation and include information on defaults and borrowing. Unpaid bills in collections represent an especially useful outcome, as they capture default on obligations such as medical and utility bills (Avery et al., 2003) and thus can provide a measure of financial distress even for the lowest-income drivers, many of whom have limited attachment to the formal financial sector.

Taking advantage of this unique panel of credit reports, I leverage staggered variation in the timing of traffic stops with an event study approach. To address the various identification concerns associated with two-way fixed effects DiD approaches raised in the recent econometrics literature (e.g., Roth et al. 2022), I estimate the event studies via the method of Callaway & Sant’Anna (2021), relying only on comparisons between individuals treated at a particular time and those treated in the future. To further mitigate concerns about violations of parallel trends, I focus my analysis on speeding violations rather than other types of infractions, such as equipment or paperwork violations, that may themselves signal changes in an individual’s financial situation. In the sample of speeders, the pretrend test of Borusyak et al. (2024) consistently cannot reject the null of parallel trends.

Event study estimates reveal that traffic fines averaging \$195 increase unpaid bills in collections by about \$34 ($se = \4). Given high payment rates on traffic fines and the fact that collections activity associated with traffic citations is very unlikely to be reported to credit bureaus, I interpret this event study estimate as a test of households’ *ex ante* ability to cover unplanned expenses, in the spirit of Dobkin et al. (2018). Specifically, this finding implies that some fraction of households resort to default on other financial obligations in order to finance the payment of a typical, unplanned expense.

This interpretation is bolstered by heterogeneity based on proxies for an individual’s financial buffer available in the credit report data. Individuals with over \$200 in available credit card balances at baseline borrow about \$19 ($se = \6) on credit cards and accrue about \$15 ($se = \4) in collections debt. On the other hand, those without easy access to liquidity on credit cards accrue an additional \$44 ($se = \7) in collections debt. Heterogeneity by both income and credit card liquidity suggests a clear hierarchy of sources for financing unplanned expenses: (i) cash-on-hand, (ii) borrowing on credit cards, (iii) delaying credit line payments, and (iv) default which ultimately leads to collections activity.

The observed increases in default following a traffic stop generate measurable, longer-term effects on access to credit. Three years out from the traffic stop, I estimate that credit scores and borrowing limits are 2.6 points ($se = 0.2$) and \$330 ($se = \50) lower, respectively. I also find evidence for longer-run declines in home ownership, geographic mobility, and attachment to the formal financial sector, proxied by whether an individual has any open credit line.

Drawing on administrative payroll records from a subset of large employers covering 20-25 percent of total employment in Florida and paying above-average wages, I find that in the

twelve quarters following a traffic stop, the likelihood that an individual appears as employed in these payroll records falls by 1.2 percentage points ($se = 0.001$), relative to a mean of 15 percent. Based on a back-of-the-envelope calculation using information in the ACS, this reduced likelihood of employment in the payroll records suggests about a \$300 decline in expected annual earnings. While fines both reduce the likelihood of transitions into the payroll records and increase the likelihood of transitions out, the impact on transitions into the payroll records is much more pronounced relative to counterfactual levels and trends. Hence, I interpret this result as evidence that fines reduce the rate at which individuals move into the subset of “good jobs” covered by the payroll records, rather than inducing transitions into non-employment, *per se*. I also find that impacts on payroll employment are wholly attributable to lower-income motorists and that those who are employed in payroll-covered jobs with above-median earnings at baseline experience no change in employment situations or financial distress following a traffic stop.

A natural question is whether the observed impacts on financial distress and job stability can be explained by other, non-fine sanctions associated with traffic tickets, such as court fees, driver license (DL) points accrued on a driver’s record and associated increases in car insurance costs, or DL suspensions imposed on non-payers. Based on imperfect data on the traffic court disposition associated with each citation, I estimate the average total financial costs of citations, taking into account the post-citation choices of motorists. My preferred estimate suggests that the average citation in my sample is associated with the payment of \$174 fines and fees initially and an \$18 increase in quarterly car insurance premiums (because many motorists avoid increases in premiums through the traffic court system), yielding a total cumulative cost estimate of about \$315 over the six quarters following the citation.

I also present estimates for subgroups of individuals based on their traffic court disposition, with the caveat that this analysis splits the sample on the post-citation choices of motorists. I find that estimates for the subgroup of individuals who can be identified as paying their fines for sure are similar to, and if anything slightly larger than, estimates in the full sample. Estimates are also comparable for a subgroup who paid their fines but avoided increases in auto insurance costs altogether, while estimates are attenuated for a subgroup who likely received fine reductions in court. No more than eight percent of the sample faced a suspension for nonpayment, and estimates are modestly larger in this subgroup. While I cannot definitely rule out a role for DL suspensions or increases in insurance costs, the available evidence suggests that fine payment itself is the primary driver of the effects.

The fact that effects on employment arrangements appear unexplained by, e.g., driver license suspensions, raises a question about mechanisms. Impacts on payroll employment are consistently strongest in subsamples with the largest increases in financial distress, suggesting a role for the impacts of a diminished credit reputation on vehicle access (e.g., [Baum 2009](#)), job-finding (e.g., [Bos et al. 2018](#); [Bartik & Nelson 2021](#)), or housing situations. My findings are also consistent with evidence that financial distress can reduce labor supply (e.g., [Dobbie](#)

& Song 2015; Barr et al. 2023) and evidence that financial distress weakens decision-making (Schilbach et al. 2016; Mullainathan & Shafir 2013) and productivity (Kaur et al., 2021). Disentangling these potential mechanisms is an interesting avenue for future research.

My central contribution is evidence that households cannot easily absorb typical, unplanned expenses. I conclude by synthesizing the key lessons from this result and discussing them in the context of the relevant literatures. First, I consider how my findings can speak to the survey evidence on the prevalence of financial fragility (e.g., FRBG 2018). Under the assumption that default is the “last resort” for covering unplanned expenses, I conceptualize the notion of financial fragility in terms of casual effects: a household is fragile if it must miss other bills to finance the payment of a traffic fine, and thus the relevant metric for comparison with the survey evidence is the share of households with positive treatment effects on collections balances. The event study estimates can identify bounds on this fraction under reasonable assumptions about the distribution of treatment effects. Based on this approach, a conservative bound implies that at least 11 percent ($se = 3$ percent), while a more data-driven estimate implies that at least 38 percent ($se = 11$ percent), of individuals borrow out of other financial obligations to cover an unexpected \$315 expense. This paper is the first to provide evidence on the prevalence of this causal notion of financial fragility.

Next, I connect my findings to the vast literature on the consumption smoothing behavior of households (e.g., Stephens 2001; Parker 2017; Ganong et al. 2020; Golosov et al. 2022; Baker & Yannelis 2017; Gelman et al. 2020; Ganong & Noel 2019). My contribution to this literature is causal evidence that default is an important consumption smoothing strategy for liquidity-constrained households, even when facing “typical” income shocks (Morduch & Schneider, 2016). My documentation of households’ tiered strategy for covering unplanned expanses is also a contribution to the literatures on low-income and behavioral household finance (e.g., Beshears et al. 2018; Gathergood et al. 2019).

My finding that a sizable fraction of households are affected by typical shocks has potentially important implications for the optimal coverage and generosity of social insurance. Specifically, the observed effects on default and the ensuing declines in creditworthiness imply that many households are not self-insured against usual income volatility. Abstracting away from the important logistical concerns, this finding implies that an expanded social safety net, which insures against a wider range of shocks faced by households, could carry social welfare gains (e.g., Mazumder & Miller 2016; Hu et al. 2019; Gallagher et al. 2019). Alternatively, the results may suggest a role for more aggressive policies to encourage self-insurance, such as expanded financial education or saving incentives programs (e.g., Klapper & Lusardi 2020, Lusardi et al. 2011).

Finally, my paper also adds to the nascent literature on the social costs of policing (e.g., Ang 2021) and a concurrent literature on the effects of legal financial obligations (LFO’s) on offender outcomes (Kessler 2020; Pager et al. 2022; Giles 2022; Finlay et al. 2022; Lieberman et al. 2023). While a large literature has examined the deterrence effects of fines (e.g.,

Makowsky & Stratmann 2011; DeAngelo & Hansen 2014; Traxler et al. 2018), interest in the potential negative effects of fines and fees on individuals has grown significantly in recent years. Finlay et al. (2022) note that an important distinction in the current research on LFO’s appears to be whether fines are coupled with criminal convictions, with studies examining variation in fine amounts among those also convicted of felonies or misdemeanors tending to find null effects on life outcomes such as reoffending or employment. My paper, on the other hand, studies comparatively small fines which are not associated with convictions and documents impacts on household financial situations.

The rest of the paper proceeds as follows. Section 2 provides the relevant institutional background and section 3 describes the data. I lay out the empirical strategy in section 4 and present results in section 5. Section 6 interprets and contextualizes the findings and section 7 concludes.

2 Institutional background

2.1 Setting

The setting for this paper is traffic enforcement in Florida. Patrolling police officers, or in some cases automated systems such as red light or toll cameras, issue citations to offenders. Traffic citations are extremely common. Over 4.5 million individual Florida drivers received at least one traffic citation over 2011–2015, with between 1.1 and 1.4 million licensed Floridians cited each year. As of the 2010 census, the population of Florida aged 18 or over was 14.8 million, implying that around 30 percent of the driving age population was ticketed at least once over this five year period.

Traffic enforcement appears to disproportionately affect disadvantaged communities. Figure 1 plots the zip code citation rate, computed as the number of citations issued to residents of a zip code divided by the number of residents, against zip code characteristics. Residents of the poorest neighborhoods are cited about twice as often as residents of the lowest poverty neighborhoods. Residents of neighborhoods with the largest minority (Black or Hispanic) populations are cited four times more often than residents of the whitest communities.

[FIGURE 1 ABOUT HERE]

2.2 Institutional details

Traffic citations specify an offense and fine to be paid. The most common violation codes over 2011–2015 were speeding (28 percent), red light camera violations (7 percent), lacking insurance (7 percent), driver not seat-belted (7 percent), and careless driving (5 percent), which account for just over half of all citations over the period. Statutory fines vary widely across offense types. For example, minor equipment violations such as broken tail-lights

carry a fine of \$110, while the fine for speeding 30+ miles per hour over the posted limit in a construction or school zone is \$620. Sanctions for certain criminal, rather than civil, traffic offenses can exceed \$1,000 and may include jail time. As discussed in section 4, I focus my analysis on speeding violations, with fines ranging from \$123 to \$273 ($\mu \approx \$195$).

Many offenses also result in “points” on a driver’s license. State law dictates that drivers accruing 12 points in 12 months (18 points in 18 months; 24 points in 36 months) have their licenses suspended for 30 days (6 months; one year). Speeding offenses are associated with 3-4 points, while points are generally not assessed for non-moving violations. In the main analysis, I focus on individuals facing their first citation in at least one year, minimizing the risk that individuals are in position to receive a points-based suspension.

Insurance companies typically consider license points as a signal of driver risk when setting premiums, so individuals may face increases in auto insurance costs following a citation. I estimate that a typical speeding citation over this period is associated with an increase in annual (monthly) auto insurance premiums of \$227 (\$19). As described below, motorists can mitigate their point exposure through the traffic court system and, taking into account those choices, I estimate that the average citation in my analysis sample is associated with a \$93 (\$8) increase in annual (monthly) premiums. See appendix C for details.

Once a citation has been issued, a driver can either submit payment to the county clerk or request a court date to contest the charge. For those contesting their citation in court, a judge or hearing officer typically decides to (i) uphold the original charge, (ii) reduce the sanctions, or (iii) dismiss the citation. A court fee, averaging about \$75, is required for those bringing their case to court, but may also be waived in some instances. For those not contesting the charge, payment is due thirty days from the citation date.¹ At the time of payment, a driver may also elect to attend traffic school. A voluntary traffic school election, coupled with an on-time payment, wipes the citation from the driver’s record and thereby prevents the accrual of the associated license points on the individual’s DL.² If the county clerk has not received payment in-full within 30 days, the individual is considered delinquent and their license is suspended, effective immediately. Knowingly driving with a suspended license is a misdemeanor offense and typically results in a fine exceeding \$300, as well as potential jail time. Figure A-1 succinctly illustrates the driver’s potential decision tree and

¹As of 2022, a new Florida law requires that counties offer income-based payment plans for traffic citations. However, during my sample period (2011–2015), only two counties, Hillsborough and Pinellas, offered three-month payment plans for traffic fines ([statute](#); [news article](#)). Figure F-2 offers suggestive evidence of smaller effects on unpaid bills in these counties during this period.

²Individuals seeking to prevent point accrual following standard non-criminal moving violations take the Basic Driver Improvement Course. The course is four hours of instruction, cannot be completed in one sitting, costs \$25 (but is typically coupled with a \$15 fine reduction), and is available online. Individuals can only complete traffic school once in any twelve-month period and five times total. About 25 percent of individuals in the subset of the main sample with valid traffic court disposition information participate in traffic school.

corresponding outcomes for the case of a typical moving violation.

If a citation remains unpaid after 90 days, the county clerk adds a late fee to the original amount owed and sends the debt to a collections agency, who then solicits payment. Collections agencies are authorized by state law to add a 40 percent collection fee to the original debt. Note that, to the best of my knowledge, collections activity originating with unpaid citations *will not appear on a driver’s credit report*.³

An important takeaway from a careful consideration of the institutional details is that the exact “treatment” a motorist faces can take many forms. Even holding the offense constant, a citation’s outcome depends on an offender’s ex-post decisions, and to some extent driving history, neither of which is perfectly observed in the data. For reasons discussed further in section 4, I focus my analysis on speeding violations, which are not associated with mandatory court appearances or automatic license suspensions, and think of the treatment as a bill for \$195 (on average), where the punishment for nonpayment is a revocation of driving privileges. But treatment could also entail time in court and court fees for those contesting their citations, increases in car insurance premiums for payers, and license suspensions for non-payers. According to the Florida Clerks and Comptrollers, who estimate that over 90 percent of traffic fines are paid on time, the threat of license suspension provides a strong payment incentive. As such, I initially focus on estimating “intent-to-treat” effects, proceeding under the assumption that fines are paid in the majority of cases. Later in the paper, I leverage traffic court dispositions data to quantify the average total costs associated with citations and present heterogeneity analyses to assess the relative importance of channels other than fine payment in explaining the effects.

It is worthwhile to consider to what extent traffic fines are similar to and different from other expense shocks faced by households. Continuing to think of a traffic fine as a bill where the punishment for non-payment within thirty days is the revocation of driving privileges, potentially distinctive features of the setting are the short, non-negotiable due date and the swiftness and severity of punishment for non-payment. Default on medical bills, for example, is quite common (e.g., Dobkin et al. 2018; Kluender et al. 2024), and the consequences for such default will generally play out over a longer time horizon in the form of medical collections appearing on credit reports and, as a result, declines in access to credit.

Given the sanction for fine non-payment, a natural parallel is unexpected car repairs, which are the most common form of financial shock faced by households according to Pew Charitable Trusts (2015). Acute or debilitating car issues in particular impose a similar tradeoff for households, where one can either pay for repairs or stop driving. On the other

³The reporting of collections activity to credit bureaus varies across both agencies and clients. I compiled a list of collections agencies used by the five largest counties in Florida by examining county clerk webpages and contacted each one directly to inquire about their reporting behavior. While most signaled an ability to report to credit bureaus on their webpage, the two agencies responding to my inquiry indicated that they do not report traffic citation-related collections.

hand, less urgent repairs highlight a contrast with traffic fines, as households can potentially defer paying for these repairs until a more financially convenient time.

Other potential parallels could include surprise family expenses (e.g., funeral costs), surprise utility bills (e.g., due to unseasonably cold weather), necessary home repairs (particularly for renters, who may be charged by landlords and then face eviction in the case of non-payment), or unexpected medical prescriptions (where a co-pay is typically required at the pharmacy). Expense shocks where households can easily reduce their exposure after the fact, adjust the timing of payment, or where default consequences accrue slowly and/or uncertainly, potentially differ from traffic fines in ways such that we might expect somewhat different household responses.

To the extent that my analysis delivers the causal effect of unexpectedly *paying* a traffic fine, as much of evidence will ultimately suggest, the traffic fine setting may also generalize to a wide array of short-run, unexpected income fluctuations, such as service industry workers receiving fewer than expected tips or shifts in a given month (Morduch & Schneider, 2016).

3 Data

3.1 Citations data

The Florida Clerks and Comptrollers Office provided administrative records of all traffic citations issued in Florida from 2010–2015 from Florida’s Uniform Traffic Citations (UTC) database. These records include the date and county of the citation, information on charged violations, and information listed on the motorist’s driver license (DL): name, date of birth, address, race, and gender, as well as the driver license state and number.

3.2 Credit reports

Access to monthly credit reports from January 2010 through December 2017 was granted by one of the major credit bureaus. I provided the credit bureau with a list of 4.5 million Florida residents issued a traffic citation between January 2011 and December 2015. Using a proprietary fuzzy linking algorithm, the driver information was matched with the credit file using name, date of birth, and home address on the citation. About 3.7 million drivers were matched to the credit file, and I further require that individuals are on file as of January 2010, have a non-missing credit score as of that date, and are aged 18-59 as of that date for analysis. I sometimes refer to this sample of 2.6M individuals as the “drivers on file” or the “initial sample.” For further information on the credit file match, see appendix E-2.

The credit bureau data represent a snapshot of an individual’s credit report taken on the final Tuesday of each month. These data include information reported by financial institutions, such as credit accounts and balances, information reported by collections agencies, information culled from public records, and information computed directly by the credit

bureau, such as credit scores (VantageScore[®] 3.0). As described in appendix E-4, I augment the credit report data by constructing an estimated income measure at baseline using an income estimate provided by the credit bureau, the average income in a motorist’s home zip code, and payroll records for a subset of the sample (described below).

For the empirical analysis, I aggregate the credit report data from the individual $\times$ month level to the individual $\times$ quarter level. This aggregation makes the dimensionality of the panel datasets more computationally manageable, with the additional benefit of reducing the (already very rare) prevalence of missing values.

3.3 Outcomes and interpretation

While credit report data provide a wealth of information on an individual’s financial situation, a challenge in working with these data is to focus on a parsimonious set of outcomes with a reasonably straightforward interpretation.

As my primary outcome, I focus on collections activity on credit reports, which represents unpaid bills that have been sent to third-party collections agencies, who attempt to recover payment. To the best of my knowledge, as mentioned in section 2.2, unpaid traffic fines will not appear as collections on credit reports. Collections are an especially useful measure of financial distress in the current context because unpaid bills need not be related to credit lines. According to Avery et al. (2003) and FRBNY (2018), only a small fraction of third-party collections originate with credit accounts; the majority are associated with medical and utility bills. Hence, unpaid bills in collections can capture increases in financial strain even among those with tenuous credit usage, whereas individuals need to maintain open borrowing accounts in order to exhibit delinquency, for example, in the credit file.

Where relevant, and in the appendix, I also show results for other measures of default such as credit line delinquencies and derogatories (e.g., accounts with a charge-off). While these primary default measures are stocks, I additionally construct a binary flow measure that equals one if an individual has any new collection, delinquency, or derogatory appear on their credit report in a given quarter.

Importantly, when examining these primary default measures, we should expect to see effects (if any) materialize gradually over time, due both to how the outcomes are defined and to the credit reporting process. For a collection to appear on a credit report, a household needs to miss a bill, a creditor needs to send that default to a third-party collections agency, and that third-party collections agency needs to report that activity to a credit bureau. In many cases, creditors (e.g., utility providers) provide some temporary forbearance on late payments before sending the debt to a collector. Hence, collections activity appearing quickly following a traffic stop could correspond to bills which have already been missed but then transition into “late enough” for the creditor to send to a collector. On the other hand, a new missed bill immediately following a traffic stop should take several months to

appear as a collection on a credit file.⁴ Note that, in either case, we should still interpret the collections activity as attributable to the fine. The same logic also applies for delinquent or derogatory credit accounts, where fines may induce already delinquent accounts to pass the threshold for reporting or lead to new defaults which ultimately become 90-days delinquent (or sufficiently late to warrant a charge-off) and then show up on a credit report.

I also explore borrowing on credit cards. In Florida, online payment of traffic fines via credit card is permitted (see figure E-1), and hence we might expect to observe impacts on credit card balances directly attributable to fine payments. Alternatively, households may offset the payment of fines by borrowing on credit cards to finance other expenses. A complication associated with credit card borrowing as an outcome in this setting is that borrowing may be constrained by credit access, which will tend to be affected by the default measures I examine. A consistent feature of the estimates for credit card borrowing is a short-run increase followed by a long-run decline, and I show that this pattern can be explained largely by a reduction in borrowing limits in the medium-term.

I interpret effects on credit card and collections balances as the extent to which households borrow, either through formal channels or by “borrowing” out of other financial obligations in the case of collections, in order to cover a traffic fine. A challenge in relying on credit report data is that default outcomes in particular may not have a clear interpretation in terms of welfare. Morduch & Schneider (2016), for example, highlight missing bills and delaying bill payments as an important consumption-smoothing tactic for cash-strapped households.

However, default can be associated with significant costs. Pattison (2020) documents that incidences of financial distress typically coincide with, rather than substitute for, declines in consumption. Moreover, dynamic consequences of default in terms of creditworthiness can be severe. A typical default instance can reduce credit scores by as much as 30 points (see figure A-13), with implications for interest rates and borrowing limits, as well as apartment leasing or job-finding. Liberman (2016) finds that such credit constraints can have significant welfare implications, estimating a typical willingness to pay of 11 percent of monthly income for a clean credit reputation. To the extent possible, I directly examine these longer-term consequences by studying impacts on credit scores and borrowing limits over a 3-year horizon.

3.4 Payroll records

Access to monthly payroll records for a subset of large employers was also provided by one of the major credit bureaus. The payroll records are quite thin and include no information on occupations or employers, but do provide earnings in each month for the subset of individuals working at a payroll-covered employer. I rely on these payroll records to explore whether

⁴Event studies in Dobkin et al. (2018) show that the impact of hospital admissions on medical collections materialize over 18-24 months. In appendix B, I show that the dynamic effects of separation from a payroll-covered job (described below) exhibit a similar pattern.

unplanned shocks can impact employment arrangements.

In my analysis sample of cited drivers, about 12-15 (16-18) percent of motorists have earnings in the payroll records in a given quarter (year). To better understand what these data capture, appendix B compares summary statistics from the payroll records with information on employment and earnings in the ACS microdata (Ruggles, 2023). Based on the ACS, the employment rate for a comparable sample of Floridians over this period was between 68 and 72 percent, suggesting that the payroll records cover about 20-25 percent of total employment in the state. For those in the payroll records, annualized earnings are about 25 percent higher than in the average job held by a demographically comparable sample in the ACS, consistent with existing evidence that larger firms pay higher wages (e.g., Brown & Medoff 1989; Cardiff-Hicks et al. 2015). I also find, via event studies, that transitions out of the payroll database are followed by increases in financial distress.

Hence, while the low coverage of the payroll records implies that transitions in and out of the payroll records do not necessarily correspond to transitions in and out of employment, the available evidence suggests that working in a payroll covered job captures something meaningful. At the very least, changes in the likelihood that a driver works in a payroll-covered job (which I term *payroll employment* when presenting the results) suggest an elevated rate of job transitions, which I interpret as evidence of employment instability.

Another benefit of the payroll records is that they provide a true income measure for a subset of the sample. As described in appendix E-4, I use the payroll information to construct an estimated income measure at baseline for the full sample and then use that measure to explore heterogeneity by income. I also present results focusing only on the subset of motorists in the payroll records at baseline, splitting that sample by earnings.

4 Empirical strategy

4.1 Event study approach

I leverage the variation in the timing of traffic stops for identification with an event study approach. Specifically, letting i index individuals and t index calendar time (in quarters), I estimate equations of the form:

$$Y_{it} = \sum_{\tau} \alpha_{\tau} + \phi_i + \kappa_t + \epsilon_{it} \quad (1)$$

where $\tau = t - \tilde{t}_i$ indexes “event time,” with $\tilde{t}_i$ denoting individual i ’s treatment timing, which I refer to as a their “cohort” (Sun & Abraham, 2021).

Of course, a wave of recent econometric scholarship has documented the various empirical issues associated with estimating event study models with two-way fixed effects (TWFE) via OLS (e.g., Chaisemartin & D’Haultfoeuille 2020; Goodman-Bacon 2021; Sun & Abraham 2021; Callaway & Sant’Anna 2021; Borusyak et al. 2024; Roth et al. 2022). Some of the

important concerns raised in this literature include the contamination of treatment effect estimates created by comparisons between currently treated and previously treated units and underidentification problems in fully dynamic specifications with no untreated group. To address these issues, I estimate the event studies using the method of [Callaway & Sant’Anna \(2021\)](#). Their approach is to construct estimates for each cohort and time period, based only on comparisons between each cohort and those treated in the future, and then aggregate these cohort $\times$ period effects into event study parameters.

Estimated via the [Callaway & Sant’Anna \(2021\)](#) approach, the event study design relies only on comparisons between individuals treated in period t and those treated in future periods. Hence, identification relies on the following parallel trends assumption: in expectation, following a traffic fine, individuals would have trended similarly to those fined in the future, had they not been stopped at that date. The key threats to this identifying assumption are differing trends between treatment and comparison groups even prior to treatment and other, unobserved shocks which coincide with treatment timing. To test for potential violations of parallel trends prior to treatment, I adopt the strategy of [Borusyak et al. \(2024\)](#). Specifically, I regress the outcome on a set of pre-treatment event time indicators, as well as individual and time fixed effects, using only the sample of not-yet-treated observations, and perform a joint significance test of the event-time indicators. I use the first four pre-treatment quarters as the time horizon for this pretrends test, because at least four quarters of pre-treatment data are observed for each cohort, and report p -values from this test.

There are two important identification concerns that bear mentioning here. First, many types of traffic infractions could signal changes in financial distress *ex ante*. For example, a citation for a broken tail-light or expired registration could be induced by a deteriorating financial situation. For this reason, I focus the event study analysis on speeding violations. Figure A-2 compares the pre-stop trends in financial distress, estimated via the [Borusyak et al. \(2024\)](#) approach, for speeding violations and non-moving violations. For non-moving offenses, the majority of which are paperwork or equipment infractions, a strong pre-citation trend ($p < 0.001$) in financial distress is evident. On the other hand, there is no such trend for speeding violations ($p > 0.35$), suggesting that the precise timing of a speeding stop is unrelated to changes in an individual’s financial situation.⁵

Second, a traffic citation of any type could signal a change to an individual’s driving patterns. There is some evidence to support this concern in the data. As shown in panel (a) of figure A-5, the likelihood that an individual has an open auto loan increases by about one percentage point in the six quarters prior to a traffic stop. On one hand, this is an important concern, suggesting that a car purchase, which could signal other changes in an individual’s situation, sometimes directly precedes a traffic fine. On the other hand, pre-stop

⁵Consistent with the evidence presented in figure A-2, figure A-3 documents that the relationship between neighborhood poverty rates and citation rates depicted in figure 1 is much weaker, and in fact reversed, when examining the citation rate for speeding infractions.

trends in the outcomes of interest are consistently zero and the presence of an auto loan on file is an imperfect indication that an individual is actively driving: less than half of the individuals in the event study sample hold an open auto loan in the quarter of their traffic stop. Moreover, as shown in panel (b) of figure A-5, the timing of an auto purchase tends to coincide with *improvements* in an individual’s financial situation, as summarized by their credit score, suggesting that bias in the event study estimates could be towards zero.

Nonetheless, I take this concern seriously. As robustness, I estimate event studies for each car purchase timing group (e.g., individuals who first purchase cars in 2011Q2) and then aggregate up across the groups. These estimates leverage only the staggered timing in traffic stops within groups of individuals who purchase cars at the same time (and prior to their traffic stops) and are quite similar to the baseline results. Additional robustness exercises are discussed below in section 5.4.

4.2 Sample construction

Motorists included in the event study sample are drawn from the initial sample of 2.6M individuals who, as of January 2010, are aged 18–59 and have a credit report with a non-missing credit score. I start with all speeding citations attributable to this set of drivers and impose the following conditions: (i) speeding is the only violation on the citation; (ii) charged speed between 6 and 29 MPH over the limit (speeds below 6 are statutory warnings and speeds above 29 require a court appearance); (iii) motorist race is either white, Black, or Hispanic. I then select the first such stop for each individual and require that the driver has no other stops in the previous year ($N = 525,646$). Figure A-4 shows the distribution of treatment timing (“cohorts”) in the event study sample as well as variation across cohorts in salient motorist characteristics.

Column 3 of table 1 reports average baseline characteristics for the event study sample. 45 percent of motorists are female, 59 percent are white, and the average age is 36. Interestingly, the analysis sample appears positively selected relative to the full set of motorists on file: those in the event study study sample have credit scores which are 20 points higher and have about \$300 less in collections debt at baseline.

In the event study sample, about 60 percent of motorists can be identified as having paid their fines for sure (“definitely paid”) based on the traffic court disposition associated with the citation. Around 30 percent *may* have received reduced sanctions through the traffic court system (“possible lenience”) and around 10 percent *may* have faced a license suspension (“possible suspension”) due to nonpayment. See appendix C and section 6 for an expanded discussion of these definitions, as well as accompanying heterogeneity analyses.

[TABLE 1 ABOUT HERE]

5 Results

Figure 2 presents event study estimates for default outcomes. In each figure, I report the p -value from the Borusyak et al. (2024) pretrends test and the static ATT estimate (the cohort weighted average from Callaway & Sant’Anna 2021). Note that, in all cases, I cannot reject the null hypothesis of parallel trends ($p > 0.357$). To assist with interpreting magnitudes, I also report estimated post-treatment counterfactual means, which are constructed by regressing the outcome on individual and time effects using only the sample of not-yet-treated observations and averaging predictions from this regression over event time (e.g., Kleven et al. 2020, Borusyak et al. 2024).

[FIGURE 2 ABOUT HERE]

As shown in panel (a), the probability of a new default flag appearing on an individual’s credit report increases by about one percentage point in the year following a traffic stop, relative to the comparison group of motorists cited in the future. This effect represents about a five percent increase relative to a mean of 0.216. The smaller static ATT ($= 0.005$) implies a relative decline in the probability of new distress flags in the longer term.

Panel (b) presents event study estimates for collections balances, which I interpret as the extent to which households borrow out of other financial obligations to cover their fines. Collections balances increase by about \$34 in the six quarters following a traffic stop. Scaled by the average statutory fine of \$195 (which may differ from the amount paid, as discussed below in section 6.1), this finding implies that, on average, about 17 percent of fine payments are financed through “borrowing” out of other financial obligations such as utility or medical bills. Panel (c) of figure 2, along with table 2, which presents the corresponding regression estimates, also shows impacts on the number of credit lines at least 90 days past due, the number of derogatory credit lines, and the number of accounts in collections.

5.1 Heterogeneity by financial buffer proxies

Of course, we should expect the impacts of fines on default to vary by whether a household has a financial buffer to draw on. I first explore the role of access to liquidity in explaining treatment effect heterogeneity by splitting the sample into two groups based on their credit card situation at baseline: those with at least \$200 in available balances on credit cards ($N = 301,318$) and those without ($N = 224,228$), which includes both those without a credit card at baseline ($N = 175,643$) and those with maxed out credit cards at baseline ($N = 48,585$).⁶ Note that, since this cut is defined at baseline, some individuals may “switch

⁶To minimize concerns about mean reversion when constructing this sample split, I compute available balance on credit cards in each quarter, defined as the revolving limit minus the revolving balance, both summed across all revolving accounts, average across the first four quarters (2010Q1-2010Q4), and define an individual as having \$200 in liquidity if this average exceeds \$200.

groups” between the baseline period and their treatment date.

As shown in panels (a) and (b) of figure 3, this proves to be an especially salient cut of the data. Estimated impacts of fines on the probability of new default and on collections balances are about three times larger ($ATT = \$43.88$) for those without access to liquidity than those with at least \$200 in available credit card balances at baseline ($ATT = \$14.83$).

[FIGURE 3 ABOUT HERE]

As shown in panel (c), credit card balances increase by about \$19 ($se = \6) in the first quarter following a traffic stop for those with available credit card liquidity, suggesting that these households finance about 10 percent of a typical fine through credit card borrowing. In the longer-term, there is a pronounced decline in credit card balances for this group. As mentioned in section 3.3, one reason to expect such a pattern is the impact of increased rates of default on access to credit, and correspondingly, panel (d) illustrates increases in credit card utilization, defined as total balances divided by total limits.

The fact that increases in utilization are comparable for both groups, but declines in balances are attributable to the group with available liquidity at baseline, implies that this liquid group, whose default responses are significantly attenuated, experiences more dramatic declines in credit limits (as confirmed in figure A-13). This result may appear surprising but can be rationalized by institutional features of credit markets. Specifically, as shown in panel (a) of figure A-13, the relationship between credit scores and credit limits is highly convex. The decline in credit limits associated with a ten point credit score decline is about \$2,550 for those with good credit scores (> 700), but only about \$140 for those with poor credit scores (< 500). Panels (b) and (c) of figure A-13 illustrate that, accordingly, the decline in credit limits following a typical default is much more dramatic for those with higher initial credit scores. Hence, the smaller increases in missed bills for this liquid group, whose baseline credit scores are 130 points higher, can generate comparatively large declines in credit limits. A similar pattern is seen in Dobkin et al. (2018), who compare the impacts of hospital visits for those with and without health insurance, finding larger impacts on default and credit scores for the uninsured, but more pronounced declines in credit card balances and limits for among the insured, who have significantly higher baseline credit scores.

Easy access to credit card borrowing is, however, far from the only measure of a household’s financial buffer or overall liquidity. In figure 4, I further split the sample by both available credit card liquidity and by baseline estimated income (cutting at the median $\approx \$31,000$), using the income measure described in appendix E-4. This sample split results in four groups of motorists: higher income with credit card liquidity ($N = 232,230$), higher income without credit card liquidity ($N = 56,046$), lower income with credit card liquidity ($N = 69,088$) and lower income without credit card liquidity ($N = 224,328$).⁷

⁷Event study estimates for credit card borrowing in the full sample, as well as estimates for all outcomes by baseline income and liquidity status are presented in supplementary appendix F.

[FIGURE 4 ABOUT HERE]

As shown in panel (a), estimated impacts on collections are most dramatic for the low-income, illiquid group (ATT = \$49). The next highest estimate is for higher-income, low-liquidity motorists (ATT = \$32), with smaller and comparable effects in the two subgroups with at least \$200 in available credit card balances (high income ATT = \$13; low income ATT = \$20). Interestingly, panel (b) illustrates a reversal of these patterns, at least with respect to the mitigating role of access to credit, when examining delinquencies on credit lines. Here, the largest effects are for the low-income but liquid subgroup, followed by low-income illiquid, high-income liquid, and high income illiquid.

As discussed in section 3.3, a potentially important consideration here is differential rates of formal borrowing: individuals must maintain open borrowing accounts in good standing in order to attain delinquencies on their credit reports. As of one quarter prior to the traffic stop, the shares of each group with at least one open credit line with the potential to transition into delinquency are: 87 percent (high income and liquid); 43 percent (high income and illiquid); 77 percent (low income and liquid) and 43 percent (low income and illiquid). Hence, these results highlight that delinquency impacts are larger for those with lower incomes and those with a greater potential for delinquency given *ex ante* borrowing.

Panel (c) of figure 4 illustrates, unsurprisingly, that the short-term increase in credit card balances seen in the prior figure is most pronounced for the subset of motorists with lower incomes and available balances on credit cards. For this group, the one-quarter event study estimate is $\beta = \$30$, or about 15 percent of a typical fine. The comparable estimate is about half the size ($\beta = \$16$) for the subset of high-income and liquid drivers and below \$10 for both illiquid subsamples. As in the previous figure, panel (d) shows estimates for revolving utilization to confirm that longer-run declines in credit card borrowing can be explained by reduced credit access.

The patterns in figures 3 and 4 suggest a reasonably straightforward hierarchy of behavior with respect to baseline financial situation. Individuals with the largest buffer, proxied by income and available credit card balances, appear to cover the majority of the fine with cash on hand, as suggested by the low rate of credit card borrowing and minor impacts of default. Those who cannot cover the fine in cash first rely on credit card borrowing, as evidenced in particular by the borrowing patterns of the subset of low-income drivers with available credit card balances, followed by “borrowing” through delaying repayment on credit lines (i.e., delinquency). And finally, borrowing out of other financial obligations, which ultimately results in collections activity, is the “last resort” for covering unplanned expenses.

5.2 Longer-run effects

I interpret the results on default primarily as providing evidence on the *ex ante* financial situations of households: the fact that fines induce default on items such as utility bills

suggests an inability to cover a \$200 expense via cash on hand for some fraction of households. This interpretation is bolstered by the above heterogeneity analyses, which reveals a clear hierarchy of payment sources.

However, increased default does not necessarily have a clear interpretation in terms of household wellbeing. If default allows consumption smoothing (Morduch & Schneider, 2016) at minimal costs, for example, then welfare could actually be increasing as collections balances accrue. As discussed in section 3.3, the welfare consequences of default are likely to play out in the longer term in the form of tighter borrowing constraints, higher interest rates, and other consequences of a diminished credit reputation or worsened financial standing, such as difficulty securing housing or employment.

In figure 5, I present event study estimates for credit scores and borrowing limits over a three-year time horizon. This figure presents estimates from both the baseline Callaway & Sant’Anna (2021) approach and an approach based on Sun & Abraham (2021) which compares only those cited in 2011–2012 to those cited in 2015Q4. I present the Sun & Abraham (2021) results, which hold the “control group” constant, to confirm that longer-run patterns are not driven only by compositional changes in the DiD comparisons.

[FIGURE 5 ABOUT HERE]

As shown in panels (a) and (b), traffic fines are associated with a 2.6 point decline in credit scores and a 1.6 percentage point increase in the likelihood of having a subprime credit score, with both effects persisting for three years following a traffic stop. Coinciding with the credit score declines, I find that borrowing limits fall by about \$330 over three years. In figure A-12, I show that along with these longer-run declines in creditworthiness are lower rates of home ownership (proxied by mortgages), geographic mobility (proxied by whether an individual’s address was updated on the credit file), and attachment to the formal financial sector (proxied by whether an individual has any open credit line).⁸

Panel (d) of figure 5 shows declines in the likelihood of working in a payroll-covered job (“payroll employment”) beginning in the first quarter following a traffic stop and persisting in the medium-to-long term. Three years out, the estimated decline in the payroll employment rate is about 1.2 percentage points, or 8 percent relative to a mean of 15 percent. As discussed in section 3.4, this finding indicates a reduced likelihood of working for large employers paying above-average wages, but does not necessarily capture employment versus non-employment.

To help interpret this estimated magnitude, appendix B-1 provides a simple, back-of-the-envelope calculation of the expected difference in annual earnings for those who are employed in the payroll records and those who are not based on information in the ACS. Taking into

⁸The 0.6 percentage point decline in the likelihood of having a mortgage shown in figure A-12 is similar to what would be predicted given the 2.6 point decline in credit scores and the estimates in Dobbie et al. (2020), who find that that a 3.2 point increase in credit scores is associated with a 1 percentage point increase in homeownership.

account both the above-average earnings in payroll-covered jobs and the fact that those working in payroll-covered jobs are employed for sure, whereas some fraction of those not in the payroll records are unemployed or out of the labor force, I estimate this difference to be about \$25,000 on average. Hence, the 1.2 percentage point decline in the likelihood of payroll employment corresponds to an expected \$300 decline in annual earnings.

5.3 Additional payroll employment results

To further unpack the impact on employment arrangements documented in panel (d) of figure 5, I first split the sample according to baseline payroll employment status. Specifically, I estimate effects separately for individuals who are consistently employed in a payroll-covered job at baseline, defined as employed for all four quarters in 2010 ($N = 55,140$), and the rest of the sample ($N = 470,506$).

Panels (a) and (b) of figure 6 illustrate that the impacts of fines on collections balances and payroll employment are present and comparable in both of these subsamples. In particular, panel (b) implies that traffic fines both reduce the likelihood of transitions into the payroll records for those not in the payroll-covered jobs at baseline and accelerate transitions out of the payroll records for those who are “employed” at baseline. The point estimate associated with the latter effect is slightly larger (-0.007 versus -0.005), but estimates are much less precise for the considerably smaller baseline employed sample.

[FIGURE 6 ABOUT HERE]

A salient question given that the payroll records cover about 20-25 percent of total employment is the extent to which declines in the likelihood of working in a payroll-covered job reflect increased rates of non-employment. Note that while the estimates for the two groups are comparable in levels, percent effects (scaled by the means) are 5.5 times larger for those not in the payroll records at baseline. Hence, a reduction in the likelihood of transitions into the payroll records appears to be an important part of the overall story. Per the discussion in section 3.4, I view this as evidence of a reduced rate of transitions up the job ladder into a subset of “good jobs,” but not necessarily evidence of elevated non-employment.

In appendix B-1, I conduct a back-of-the-envelope calculation which combines the estimated impacts by baseline payroll status with information on typical employment transitions in the CPS from Fujita et al. (2020) to estimate an implied impact on non-employment. This exercise suggests that, if transitions in and out of the payroll records follow the same pattern as the average transitions into and out of employment at a given employer in the CPS, the static ATT estimate (-0.006) and longer-run event study estimate (-0.012) imply 0.2 and 0.4 percentage point increases in the likelihood of non-employment, respectively.

In panels (c) and (d) of figure 6, I show results using only the subsample in the payroll records at baseline and split motorists at the median of annualized payroll earnings

($\approx \$34,000$). Both panels reveal stark heterogeneity by baseline income. For “employed” motorists with below-median earnings, collections balances increase by \$82 and the likelihood of working in a payroll-covered job declines by 1.2 percentage points. For those with above-median earnings, the comparable estimates are \$20 and 0.4 percentage points, with neither effect statistically distinguishable from zero. One could view panels (c) and (d) as a reassuring placebo test for the validity of the event study approach: individuals with stable employment and above-median earnings at baseline experience no detectable change in either collections debt or the likelihood of working in a payroll-covered job following a traffic stop.

As discussed in detail below in section 6.2, I find minimal evidence that the estimated impact on employment situations can be attributed to institutional features such as DL suspensions or other increases in driving costs, as opposed to the financial shock of a traffic fine, raising the question of why paying an unplanned expense may affect employment arrangements. As shown in figure 6, declines in the likelihood of working in a payroll-covered job are concentrated among lower-income motorists, who also see the largest increases in financial distress. Hence, the hypothesis that weaker financial standing and a diminished credit reputation induces employment instability or reduces the ability to obtain or hold good jobs is at least consistent with the evidence. In particular, a lower credit score could affect job-finding directly (e.g., Bartik & Nelson 2021; Bos et al. 2018) or indirectly through a compromised ability to secure new housing, for example.

While this channel is a reasonably compelling explanation for reduced transitions into the payroll records, it has less bite as an explanation for the increased separation rate documented in panel (d) of figure 6. This pattern is, however, consistent with the result of Dobbie & Song (2015) that financial distress reduces labor supply and the finding in Barr et al. (2023) that cash transfers in a particularly financially-constrained sample increase labor supply. My results are also consistent with a growing body of work documenting the psychological costs of financial distress (e.g., Mullainathan & Shafir 2013; Schilbach et al. 2016), including lower productivity (Kaur et al., 2021). Disentangling these potential explanations is beyond the scope of this paper but presents an interesting avenue for future research.

Another possible mechanism is the impact of fines on car access or ownership. While the evidence discussed below in section 6.2 suggests a minimal role for license suspensions or insurance costs, changes in financial standing could affect households’ ability to acquire or retain vehicles (Dobkin et al. 2018; Dobbie et al. 2017), and there is evidence that vehicle access can have important implications for employment outcomes (e.g., Raphael & Stoll 2001; Baum 2009; Johnson 2006). Table B-2, which presents results from a mediation-style analysis, provides some support for this hypothesis. Specifically, I divide the sample into forty groups based on demographics and baseline income, estimate group-specific impacts on payroll employment in the longer-run and several possible mediators in the medium-run, and explore the ability of the group-specific medium run effects to explain the group-specific long-run effects on payroll employment. Estimates suggest that larger declines in the likelihood

of holding an auto loan in the medium term are a statistically significant predictor of larger declines in payroll employment in the longer-run.

One could alternatively ask to what extent the payroll employment effects can themselves explain the observed increases in financial distress. An important consideration here is dynamics: new defaults induced by changes in (payroll) employment status will typically take several quarters to accrue on a credit report, and by this logic alone, gradual changes in payroll employment status would appear quite unlikely to explain the initial increases in default. Another consideration is magnitudes. Based on the back-of-the-envelope calculation above, the longer-run effect on payroll employment implies a \$25 decline in monthly earnings. Abstracting from dynamics, this earnings change would predict a \$5 increase in collections debt based on the collections-earnings elasticity estimate in figure B-3.

5.4 Robustness

A central concern for the validity of the event study estimates is the possibility that traffic stops coincide with significant life changes that, for example, result in increased driving and also predict declines in financial situation. Bolstering this concern is the finding that car purchases, proxied with the presence of an open auto loan on the credit file, increase over the six quarters leading up to a traffic stop, as shown in figure A-5. To partially address this worry, figure A-6 reports event study estimates computed within auto purchase cohorts.

Specifically, for each individual i , I compute the first quarter in which I observe them as having an open auto loan on the credit file $\tilde{z}_i$. I then estimate event studies separately for each $\tilde{z}$ group and aggregate up the group-specific estimates, weighting by sample shares. These estimates leverage staggered variation in the timing of traffic stops only within groups of individuals who purchase cars at the same time. This exercise is similar in spirit to the procedure of [Freyaldenhoven et al. \(2019\)](#), who suggest using changes in a relevant observable to purge trends in the outcome attributable to changes in unobservables around the event.

There are two complications faced by this approach. First, many individuals (~ 20 percent) never have an auto loan in the credit bureau data. And second, because auto purchases are related to a household’s financial situation, which is the outcome of interest, one should not condition on the timing of purchases made after the traffic stop. Accordingly, I construct the auto purchase cohorts using the subgroups whose first auto purchases occur in 2010Q1 through 2012Q4 (to allow a sufficient post-event window for estimating the event studies) and whose traffic stop occurs after their auto purchase.

Figure A-6 presents event study estimates first using the baseline identification strategy but for this subsample with early auto purchases predating their traffic stops ($N = 291, 134$; 55 percent of the main sample) and then using the within-auto cohorts identification strategy for this subsample. Conditioning on the timing of auto purchases has minimal qualitative or quantitative impact on the pattern of results, with event study estimates and associated

ATT’s using the within-cohorts strategy nearly always within the 95 percent confidence interval of the estimates using the same sample but baseline event study approach.

Another concern for the validity of my approach, again potentially heightened by the slight increase in auto purchases prior to a traffic stop, is the possibility that deteriorating financial situations following a traffic stop could reflect mean reversion. While statistically insignificant pre-trends in a host of outcomes of interest and the similarity of the within-auto purchase cohort estimates to the baseline results (to the extent that auto purchases capture any improvements in financial situations leading up to a traffic stop) work against this hypothesis, I present two additional robustness checks in figures A-7 and A-8.

In figure A-7, I present estimates within *predicted* treatment cohorts. Specifically, I regress treatment timing on baseline characteristics as well as changes in auto loans and credit scores over the first year of the data. From this regression, I compute a predicted cohort for each individual, divide individuals into deciles of this prediction, and estimate event studies which rely on comparisons only within these deciles. Were mean reversion an important concern, we should expect meaningful attenuation when conditioning on predicted treatment timing based on levels and changes in financial situations. Instead, figure A-7 shows that results from this approach are very similar to the baseline estimates.

Figure A-8 examines pre-stop trends over a longer time horizon. A salient issue here is that, while all individuals contribute to the identification of the estimates for the first four pre-stop quarters (because four quarters of pre-stop data are available for everyone), estimates beyond four quarters also reflect changes in the composition of cohorts contributing to identification. To assess pretrends over a longer horizon, then, I reweight each cohort to match the average baseline characteristics of the full sample, thus ensuring that the longer-run pre-period estimates are not driven by differences in characteristics across cohorts. These cohort-reweighted estimates reveal no pre-stop trends in all outcomes of interest for a full three years prior to the traffic stop, further reducing concerns about mean reversion. Importantly, treatment effect estimates (both dynamic event study estimates and static ATT estimates) are quite similar in the baseline and cohort-reweighted specifications.⁹

⁹As additional robustness, figure A-9 shows that estimates obtained via the methods of [Sun & Abraham \(2021\)](#) or [Borusyak et al. \(2024\)](#) are nearly identical to the main estimates, figure A-10 shows that estimates are unchanged when relying instead on monthly data, and figure A-11 shows that results are not driven by trend breaks for the counterfactual control group. Supplementary appendix G presents a qualitatively similar pattern of results using a complementary instrumental variables strategy which does not exploit variation in traffic stop timing.

6 Discussion

6.1 Quantifying the shock

In terms of interpreting the empirical findings, a critical question is how sizable of a financial shock is the average traffic citation in practice. As discussed in section 2, the ultimate sanctions faced and paid by each motorist depends both on the statutory sanctions associated with an offense and on post-citation choices made by motorists, such as whether to contest a citation in traffic court. One could think of the task at hand as estimating a “first stage” by which the event study estimates should be scaled.

Using data on traffic court dispositions associated with each citation, I focus on quantifying three types of costs: fine payments, court fees, and increases in auto insurance premiums triggered by the accrual of points on a motorist’s driver license. The central challenge I face is that the disposition records only provide definitive information on a citation’s outcome in a subset of cases. In the event study sample ($N = 525,646$), 33.2 percent and 25.7 percent of citations have dispositions indicating fine payment and traffic school, respectively. For these citations, statutory fines were paid in full and there were no associated court fees. Those with paid dispositions accrue the DL points associated with their offense, while those with school dispositions do not accrue license points.

The remaining 41 percent of citations in the sample, with missing dispositions (1.8 percent) or disposition codes of guilty (6 percent), dismissed (8.7 percent), or “adjudication withheld” (24.6 percent), present significant interpretation challenges regarding the ultimate outcome of the citation. In particular, those with dismissed or withheld verdicts may have faced no sanctions, received partial penalty reductions (such as the waiving of license points), or pled down to lesser offenses. These complications, as well as my approach for constructing estimated payments in light of these challenges, are discussed in detail in appendix C.

[FIGURE 7 ABOUT HERE]

Panels (a) and (b) of figure 7 present estimates of the total costs of citations which take into account the post-citation choices of motorists and the various complications associated with mapping dispositions into ultimate payment outcomes. As shown in panel (a), I estimate that, in the quarter of the citation, the average payment of fines and court fees equals \$174. Insurance premium increases phase in over the first three quarters and reach a maximum of \$23 per quarter. This estimate reflects the fact that a meaningful share of the sample likely avoided the accrual of license points (and therefore insurance cost increases) via either traffic school or the traffic court system. Panel (b) illustrates the estimated cumulative costs, factoring in both the initial payment of fines and fees and additional insurance payments over time. My preferred estimate is that, after six quarters, the average motorist has paid \$302 in costs associated with their traffic citation.

Shaded regions in panels (a) and (b) report “confidence bands” that illustrate the sensitivity of these estimates to various assumptions about the outcomes of citations with court-related verdicts and associated insurance cost increases, described in more detail in appendix C. Upper and lower bound estimates on combined fine and fee payments are \$216 and \$114, respectively. Estimated increases in insurance costs are more sensitive to assumptions, with cumulative insurance cost estimates ranging from \$58 to \$252. Combining upper and lower bounds on each yields a range of total cumulative cost estimates from \$172 to \$562, comparable to the \$400 emergency expense considered in surveys, discussed in section 6.3.

Note that increases in insurance costs persist for three years following a citation, after which insurance companies no longer consider the offense when pricing. My middle-ground estimate of total costs paid over three years is \$443, but over this longer time horizon, I cannot rule out total cumulative costs as large as \$940 (with \$724 in total insurance payments). On one hand, these more significant long-term costs could play a role in generating the long-run effects presented in figure 5. On the other hand, the analysis below provides suggestive evidence that effects are similar even for those facing no changes in insurance premiums.

Another consideration in quantifying the total costs associated with citations in my sample is additional traffic stops following the focal citation. Figure C-5, which presents event study estimates for cumulative fines paid, suggests that the average citation is associated with an additional \$13 in traffic fines paid over the following six quarters, increasing my estimate of the total costs faced by the average motorist to \$315.

6.2 Fine payment versus other mechanisms

To shed light on the relative importance of fine payment *per se*, as opposed to other institutional features such as traffic court involvement, the accrual of license points, or license suspensions imposed on non-payers, in explaining the observed effects, figure 8 presents event studies for subsamples based on the traffic court disposition associated with the citation. While I view this exercise as descriptively useful, it is important to note that the sample is being split on endogenous, post-citation choices made by motorists.

[FIGURE 8 ABOUT HERE]

In panels (a) and (b), I show event study estimates for collections balances and payroll employment for the full sample, as well as for three subgroups: (i) those with dispositions indicating fine payment;¹⁰ (ii) those with dispositions indicating dismissal or withheld adjudication, which I call the “possible lenience” subgroup; and (iii) those with missing or guilty

¹⁰To keep the benchmark results the same throughout figure 8, the payer group in panels (a) and (b) is the group with verdict = 4, which is a subgroup of the “definitely paid” sample, as that sample also includes those who attend traffic school. Estimates for the entire “definitely paid” sample are similar to those for this group, as can be seen from panels (c) and (d) of figure 8.

dispositions, which I call the “possible suspension” subgroup. As discussed in appendix C, it is tempting to view the “possible lenience” subgroup as a placebo group, but there are several important caveats to this interpretation. This subgroup of individuals must have attended traffic court during a workday and would have faced a \$75 court fee, which may or may not have been waived regardless of the outcome of the court hearing. Moreover, individuals who receive only partial reductions in penalties via traffic court, such as a reduced charge or a waiving of license points, will show up in the disposition records as having their case dismissed or adjudication withheld. Hence, a sizable share of this group almost surely faced some form of (albeit reduced) sanctions.

Aligning well with these institutional caveats, panels (a) and (b) of figure 8 show that estimated impacts for this “possible lenience” group are significantly attenuated relative to the the estimates for fine payers, about half the size in each case, but non-zero. For the subgroup of motorists who may have faced DL suspensions due to nonpayment, the increase in collections balances is significantly more pronounced relative to the increase for fine payers. For payroll employment, however, this dimension of heterogeneity is less stark; while the point estimates are consistently more negative for the possible suspension group, the estimated overall ATT is similar in both subsamples.

Arguably the most important takeaway from panels (a) and (b) of figure 8, however, is the fact that estimates for the subgroup who can be identified as paying their fines for sure are similar, and if anything slightly larger, than estimates in the full sample. This finding suggests fine payments (and insurance costs), as opposed to license suspensions, as the primary driver of the financial distress and employment instability effects in the full sample. Importantly, this result also highlights that the main collections balances estimate cannot be explained by collections originating with unpaid citations, since those with paid fines would not be subject to collections activity associated with their citation.

In panels (c) and (d) of figure 8, I assess the relative importance of driver license points, which are accrued by fine payers and can affect future car insurance premiums, in explaining the results. Specifically, I compare effects for those with “paid” (same as above) and “traffic school” disposition verdicts. Traffic school attendees are required to pay their fines but, in return for completing a four hour course, do not accrue the driver license points associated with the citation. Hence, comparing effects for these two groups can shed light on the relative importance of accruing DL points. For both collections balances and payroll employment, estimates appear similar, but slightly smaller, for those attending traffic school.

For collections balances, these estimates indicate a modest downward trend in collections balances prior to the citation, which perhaps make sense: those with the wherewithal to opt for traffic school are also those with improving financial situations. Therefore, I also present estimates for the traffic school group, reweighting these motorists to match the characteristics of the payer sample based on baseline age, gender, race, and quartiles of credit score and estimate income. This reweighting eliminates the downward pretrend for the traffic school

group and slightly increases the treatment effect estimates. With reweighting, the effects for payers (ATT = \$43) and school attendees (ATT = \$40) are remarkably similar, suggesting a minimal role for DL points in explaining increased financial distress. Reweighting also reduces, but does not quite close, the gap in the payroll employment estimates for payers (ATT = -0.008) and school attendees (ATT = -0.006).¹¹

6.3 Financial fragility in surveys

The central contribution of my analysis is the finding that typical, unplanned expense shocks can have important implications for household financial situations. I interpret the headline event study results on collections balances as evidence that a meaningful share of the average payment of a \$300 unplanned expense must be financed through default on other financial obligations. This interpretation is supported by the analysis in section 5.1, which suggests that default ultimately leading to collections activity is the “last resort” for households and the finding from section 6.2 that default effects are comparable for the sample of payers.

Motivated by the much-cited survey evidence on the share of households self-reporting an inability to cover emergency expenses (e.g., FRBG 2018), a related question of interest is about the *distribution* of ability to cover emergency expenses. In other words, based on my estimates, what share of individuals cannot cover traffic fines without defaulting on other obligations and how does that share compare with prominent survey estimates? Writing the treatment effect of fines on financial distress for individual i as $\Delta_i = Y_i(1) - Y_i(0)$, this amounts to quantifying $\pi = Pr(\Delta_i > 0)$.

A natural first step here is to address a lingering question about the event study results. At first glance, the estimated impacts on the probability of new default events (extensive margin) and on collections balances (intensive margin) may seem inconsistent. The former suggests “small” effects of fines on the share of households with any financial distress while the latter suggests larger effects on default amounts. The key to reconciling these two results is to note the very high counterfactual mean on the extensive margin: $\mu = 0.22$. In other words, default is quite common in this sample and many households would have missed a bill regardless of whether they faced traffic fines. However, the unexpected fines induce *additional* default on the intensive margin for those who would have defaulted to some extent.

Hence, an accurate characterization of the fraction of households affected by fines requires consideration of both the extensive and intensive margins. The share of households induced to borrow from other financial obligations depends on the full distribution of treated and untreated potential outcomes and is therefore unidentified by the event studies (Borusyak,

¹¹Figure F-1 provides additional evidence that initial fine payment, as opposed to downstream increases in auto insurance costs, is the primary driver of the effects by showing that results are similar for those cited near and far from their home county. Those cited near home are more likely to have dismissed/withheld verdicts instead of paid verdicts, translating to a small reduction in estimated paid fines but relatively large reductions in insurance cost increases.

2015). However, one can place ad-hoc bounds on this notion of π by making assumptions about the distribution of treatment effects.

The average treatment effect I estimate can be expressed as $\hat{\Delta} = \sum_{\Delta=\underline{\Delta}}^{\bar{\Delta}} \Delta \pi_{\Delta}$, a weighted average of all possible treatment effects $\Delta \in [\underline{\Delta}, \bar{\Delta}]$, with weights π_{Δ} equal to the share of the sample with treatment effect Δ . I assume that treatment effects are bounded below by $\underline{\Delta} = 0$; in other words, no one is made *less* likely to default by an expense shock. Hence, the average treatment effect can be written as $\hat{\Delta} = \sum_{\Delta=0}^{\bar{\Delta}} \Delta \pi_{\Delta} = \sum_{\Delta>0}^{\bar{\Delta}} \Delta \pi_{\Delta}$ and the goal is to bound $\pi = \sum_{\Delta>0}^{\bar{\Delta}} \pi_{\Delta}$, the share with positive treatment effects. To see that a sharp lower bound on π is identified by $\bar{\Delta}$, the maximum treatment effect size, note that:

$$\underline{\pi} = \min_{\pi_{\Delta}} \sum_{\pi>0}^{\bar{\Delta}} \pi_{\Delta} \text{ s.t. } \sum_{\Delta=0}^{\bar{\Delta}} \pi_{\Delta} \Delta = \hat{\Delta}$$

yields $\underline{\pi} = \hat{\Delta}/\bar{\Delta}$, the ratio of the average and maximum treatment effect.

A useful starting point, then, is to assume $\bar{\Delta} = \$315$, the estimated average cumulative cost of a citation, including insurance premiums and future fines, as discussed above in section 6.1. In other words, no household is induced to default on more than \$315 by a \$315 expense shock. Based on the estimated ATT on collections balances ($= \$34$), this implies a lower bound on π given by $\underline{\pi} = 34/315 = 0.108$ ($se = 0.03$).

This lower bound is likely too conservative. On average, fines also induce default on credit lines, suggesting that defaults which lead to collections are not the only margin of adjustment. Further, there is no evidence for treatment effects as large as \$315 in the data; the estimated ATT for the low-income, illiquid group in figure 4 is \$48.62 ($se = \9). In appendix D, I estimate an upper bound on treatment effects using an approach which first bins the data into cells based on age, gender, race, as well as baseline estimated income, credit score, collections debt, and available credit card liquidity. I then use heterogeneity in treatment effects based on these characteristics in one half of the data to predict heterogeneity in the other half, estimate effects by deciles of these predictions, and extrapolate to the implied estimate for the individual with the highest predicted treatment effect. This approach yields an estimated $\bar{\Delta} = \$89$ ($se = \$23$), with an associated $\underline{\pi} = 0.383$ ($se = 0.109$).

As robustness, I supplement this extrapolation-based approach with a companion strategy which estimates treatment effects for covariate groups and uses extreme quantiles of the deconvolved (i.e., net of estimation error) distribution of treatment effects as an estimate for $\bar{\Delta}$. This approach gives magnitudes ranging from \$72 to \$112 ($se = \$17$), with the largest of these estimates implying that $\underline{\pi} = 0.302$ ($se = 0.059$).¹²

One can also estimate an upper bound on π by examining the share of individuals who are induced into their *first ever* default by fines. Panel (a) of figure A-14, which presents event

¹²See appendix D for additional details on the estimation of $\bar{\Delta}$, including a graphical depiction of the extrapolation-based estimate, deconvolution-based estimates for various definitions of the covariate cells, and additional information on the computation of standard errors for $\bar{\Delta}$ and $\underline{\pi}$.

study estimates where the outcome is an indicator for whether a motorist has accumulated any new default flag to date, implies that about 0.5 percent of the 39.4 percent of the sample at risk of being “pushed” into their first default by traffic fine are actually induced to default, suggesting that at least 38.9 percent of the sample is unaffected by fines.¹³

To summarize, conservative bounds suggest that between 11 and 61 percent of households are induced to default on other financial obligations by an unplanned, \$315 expense. My preferred estimate, which relies on an extrapolation-based estimate of the maximum treatment effect, suggests that at least 38 percent ($se = 11$ percent) are induced to default. An alternative approach for estimating the maximum treatment effect yields similar conclusions.

How do these bounds compare to existing evidence on this notion of financial fragility? While the most heavily-cited statistic is the finding from the 2017 Survey of Household Economics and Decisionmaking (SHED) that only 60 percent of households would cover an emergency \$400 expense with cash (FRBG, 2018), the same survey includes another question which is more directly comparable to my analyses. Specifically, 15 percent of respondents indicated that they would miss other monthly bills if faced with an emergency \$400 expense (Reynolds 2019; Strain 2019). As a counterpoint to the SHED, Bhutta & Dettling (2018) use the Survey of Consumer Finances (SCF) to estimate that about 24 percent of households could not pay a \$400 expense using liquid savings.

Importantly, the composition of my sample of traffic offenders may differ from that of a nationally representative survey. Hence, in table 3, I present estimated ATT’s for collections balances and implied lower bounds on π when reweighting my sample to match the demographic, neighborhood income, and credit score distributions in Florida and U.S. Compared with my baseline $ATT = \$34$, reweighted estimates range from \$27.7 to \$30.1, suggesting that the inability to meet unplanned expenses is marginally more pronounced in my sample than in the broader population. Associated estimates of π are around 0.09 when using the conservative $\bar{\Delta} = \$315$, around 0.25 when using the largest deconvolution-based $\bar{\Delta} = \$112$, and around 0.32 when using the extrapolation based $\bar{\Delta} = \$89$.

[TABLE 3 ABOUT HERE]

These reweighted estimates of the lower bound affected share (π), which suggest that at least 25 percent of households are induced to default by a traffic fine (when using the data-driven estimates of the treatment effect upper bound), are larger than would be expected based on the SHED (15 percent) and comparable to what would be expected based on the SCF (24 percent). Hence, I view my estimates as roughly consistent with the survey evidence but suggestive of higher rates of this notion of financial fragility.

¹³This finding suggests an important dimension of heterogeneity in the data, further explored in panel (b) of figure A-14, which shows no effects of fines on collections balances for those with no default flags on their credit report at baseline (38 percent of the sample).

Note that we may not necessarily expect the causal evidence I present to align with the survey evidence. In the SHED, household expectations may differ from reality, and one could read my findings as suggesting that households overestimate their financial resiliency. The [Bhutta & Dettling \(2018\)](#) estimate from the SCF could overestimate the share of households that would actually be affected by a \$400 expense if some households have easy access to other sources of liquidity, such as loans from family. On the other hand, [Chen \(2019\)](#) has pointed out that their estimate is sensitive to various computational and modeling assumptions.

6.4 Relevance beyond the United States

The discussion above implies a non-negligible prevalence of financial fragility, or the inability to cope with typical, unplanned expenses, in the United States. A natural follow up question is: to what extent is this also a salient issue elsewhere in the developed world?

The EU Statistics on Income and Living Conditions (EU-SILC) project conducts a yearly household survey which asks a comparable question to that in the SHED about whether a household could manage an unexpected expense using its own resources. The magnitude of the expense in the EU-SILC survey varies across countries and is defined as five percent of the country’s median personal income, which is one month’s income for a household at the poverty threshold. Hence, the magnitude of (hypothetical) unexpected expenses examined in the EU-SILC is significantly larger as a fraction of typical incomes than the \$400 figure in the SHED, which is about one percent of median personal income in the U.S.

Averaged across the E.U., the share of households reporting an inability to cover emergency expenses is 32 percent, less than the comparable figure in the U.S. based on the SHED (39 percent). Factoring in the stark differences in the magnitude of the expense shocks considered (\$987 in the EU-SILC versus \$400 in the SHED), it may be reasonable to conclude that this notion of financial fragility is moderately to significantly less prevalent in the E.U.

One could also benchmark the size of the emergency expenses considered in the EU-SILC with the magnitude of the typical speeding fine. Average speeding fines in the E.U. are marginally higher than in the U.S. as a share of monthly incomes (8 percent versus 5 percent). However, speeding fines correspond to a significantly smaller share of the expense shock considered in the EU-SILC (14 percent) than in the SHED (37 percent). Combined with the higher self-reported rates of ability to cover larger expenses, then, a higher fraction of households are likely able to weather these fines in the E.U. than in the U.S. For further details, see table A-1, which reports the size of the expense shock, share of households reporting an inability to cover that shock, as well as information on typical speeding fines for 25 E.U. countries, as well as population-weighted averages.

Differences across European countries in the prevalence of self-reported financial fragility could be instructive in terms of understanding differences between the U.S. and the developed world more broadly. There is substantial heterogeneity in the share of households

reporting fragility across countries, with rates exceeding 50 percent in Croatia, Greece, and Latvia and below 20 percent in Norway. Demertzis et al. (2020) note that potentially important determinants of differences across countries in this regard are differences in financial literacy and differences in culture, specifically as it relates to relying on families, friends, and communities for financial assistance (Lusardi et al., 2011).

Another potential determinant of differences across countries could be the generosity of the social safety net. Figure A-15 shows that a simple proxy for the generosity of a country’s tax and transfer system, the difference in pre- and post-tax poverty rates, can explain about 25 percent of the variation in self-reported financial fragility. Hence, overall rates of redistribution could partially explain differences in financial fragility between the U.S. and Europe. Note that this conclusion meshes well with an emerging literature documenting the role of social insurance programs in the U.S. in insuring households against financial distress (e.g., Mazumder & Miller 2016; Gallagher et al. 2019).

6.5 Consumption smoothing and social insurance

A vast literature in economics has examined the consumption responses to income fluctuations (e.g., Stephens 2001; Parker 2017; Ganong et al. 2020; Golosov et al. 2022; Baker & Yannelis 2017; Gelman et al. 2020; Ganong & Noel 2019). A contribution of my paper to this literature is compelling evidence on the consumption smoothing strategies of household when faced with typical negative income shocks. The evidence presented in section 5.1 implies a “pecking order,” with households first drawing on cash-on-hand, followed by borrowing on credit cards, followed by delinquency on credit cards, and finally default on other obligations which ultimately results in collections. More broadly, the evidence suggests that default is an important consumption smoothing strategy for liquidity-constrained households. While surveys and ethnographies have suggested the prevalence of this behavior (e.g., Morduch & Schneider 2016), my analyses confirm this by showing that typical shocks causally induce default using a large panel of individuals.

From the perspective of a policymaker, the consumption smoothing behavior of households is particularly relevant for the optimal generosity of social insurance programs. In canonical models (e.g., Baily 1978; Chetty 2006), more generous social insurance benefits are socially desirable as the difference in a household’s marginal utility of consumption between the “bad” and “good” states increases, holding constant moral hazard effects. If households can self-insure, there is no need for social insurance.

My empirical findings speak to the differences in marginal utilities associated with typical, negative income shocks in a dynamic sense: shocks induce default, which in turn reduce future welfare through tighter borrowing limits, higher interest rates, and employment instability, as discussed in section 5.2. While quantifying the implied differences in marginal utilities is beyond the scope of this paper, the results suggest the important lesson that many

households are not self-insured against even typical income fluctuations. Hence, an expanded social safety net which insures against a larger set of usual shocks could yield social welfare gains, abstracting away from the obvious logistical and moral hazard concerns.

Alternatively, the results may imply a need for policy interventions to encourage self-insurance, such as financial education or savings incentives (e.g., [Klapper & Lusardi 2020](#), [Lusardi et al. 2011](#)). The relative social welfare gains from an expanded social insurance system versus expanded financial education, for example, would depend on the moral hazard costs of insuring against a wider range of shocks and on the relative effectiveness of, e.g., the financial literacy program.

7 Conclusion

Motivated both by a growing body of evidence suggesting the inability of low-income households to cope with unexpected expenses and the observation that the incidence of policing falls largely on disadvantaged communities, this paper studies the effect of fines for speeding violations on financial wellbeing. To estimate causal effects, I link administrative traffic citation records to a panel of credit reports for cited motorists and rely on the staggered timing of traffic stops for identification.

I find that traffic fines averaging \$195 and associated with payments of \$315 over the following six quarters are associated with increases in unpaid bills in collections of about \$34. I interpret this as evidence that some fraction of households must default on other financial obligations in order to cover unplanned expense, which that interpretation supported by two additional pieces of evidence. First, the effect of fines on unpaid bills appears attributable to fine payment itself, rather than other institutional explanations such as traffic court involvement, driver license suspensions, or the accrual of driver license points. And second, heterogeneity analysis by proxies for an individual’s financial buffer reveal that default which ultimately leads to collections activity is the “last resort” for households, preceded by paying with cash on hand, formal borrowing, and delaying credit line payments as sources for financing unexpected expenses.

In turn, increased default leads to measurable longer-run effects on access to credit, with credit scores and borrowing limits falling by 2.6 points and \$330, respectively, over the three years following a traffic stop. I also find evidence that this worsening financial position is associated with a 1.2 percentage point (eight percent) decline in the likelihood of appearing as employed in a database of payroll records from large employers. I interpret this finding as suggestive of a diminished ability to obtain or hold “good jobs.”

The evidence that fines are associated with increased financial distress, declines in credit reputation, and changes in employment situations suggests that many households cannot easily absorb typical, but unplanned, expense shocks. A conservative estimate implies that at least 11 percent ($se = 3$ percent), while my preferred approach implies that at least 38 percent

($se = 11$ percent), of households are induced to borrow from other financial obligations when faced with an emergency \$315 expense, consistent with recent survey evidence on the prevalence of financial fragility in the United States.

Data and code availability statement

The analyses in this paper rely on a combination of publicly available data and confidential data which cannot be shared. All publicly available data and all programs used to produce the results in this paper are provided in a Zenodo repository at <https://dx.doi.org/10.5281/zenodo.13845033>. The README file included in this repository also provides instructions for obtaining the confidential data used in this paper. To the extent possible, the author will assist with any reasonable replication attempts for one year following publication.

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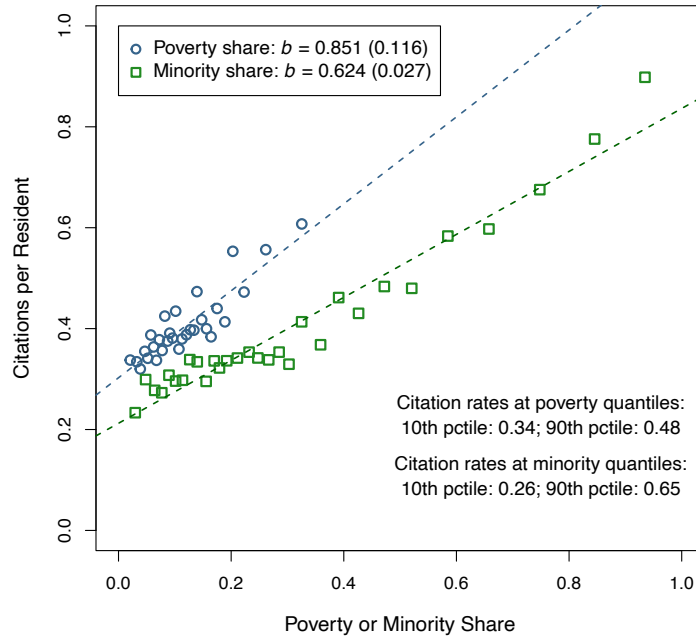
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Figure 1: Citation rates by neighborhood characteristics



Notes: This figure plots binned means of the neighborhood ticketing rate (total citations 2011–2015 issued to zip code residents divided by the number of residents) against binned means of neighborhood characteristics. $N = 908$ zip codes.

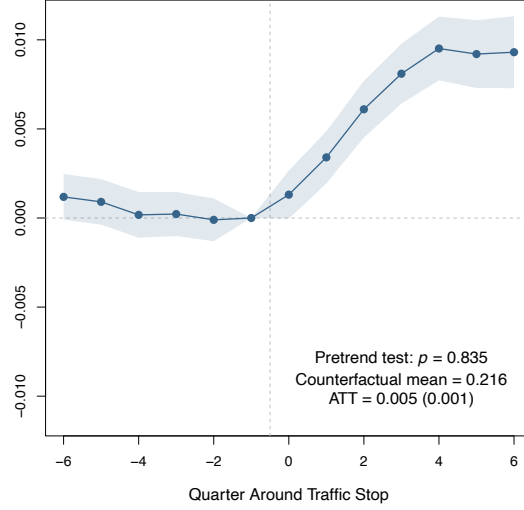
Table 1: Summary statistics at baseline

	(1) Florida	(2) Drivers on File	(3) Event Study
<i>Panel A: Demographics</i>			
Female	0.51	0.44	0.45
Race = White	0.55	0.4	0.59
Race = Black	0.16	0.17	0.2
Race = Hispanic	0.24	0.22	0.22
Age	39.3	36.81	36.37
Credit File Age	—	13.02	13.2
Credit Score	662	604	624
Estimated Income	32000	35137	39524
Zip Income	52874	51481	55023
<i>Panel B: Financial Distress</i>			
Collections		2.83	2.24
Collections Balances		1636	1299
Delinquencies		2.21	1.99
Derogatories		1.62	1.43
<i>Panel C: Credit Usage</i>			
Any Revolving		0.66	0.73
Any Auto Loan		0.36	0.41
Any Mortgage		0.28	0.33
Revolving Balances		4023	4950
Revolving Limit		12177	15367
<i>Panel D: Payroll Records</i>			
Any Payroll Earnings		0.12	0.13
Monthly Earnings		2975	3319
<i>Panel E: Citation Information</i>			
Fine Amount		171.85	195.53
DL Points		1.74	3.39
Definitely Paid		0.465	0.589
Possible Lenience		0.401	0.333
Possible Suspension		0.134	0.078
Individuals	14800000	2631641	525646

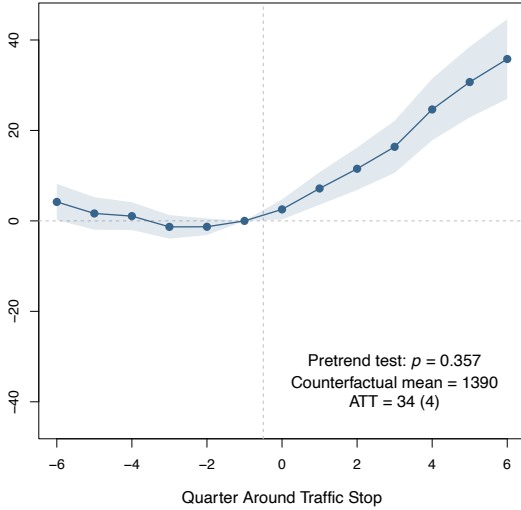
Notes: This table reports summary statistics as of 2010Q1 across samples. Column 1 reports statewide means computed from the ACS or provided by the credit bureau. Column 2 reports means for the “initial sample” of drivers who are (i) matched to the credit file, (ii) present on the credit file as of 2010Q1, and (iii) aged 18-59 and have a credit score as of that date. Column 3 reports means for the event study sample. See text for additional details on sample construction.

Figure 2: Event study estimates for default outcomes

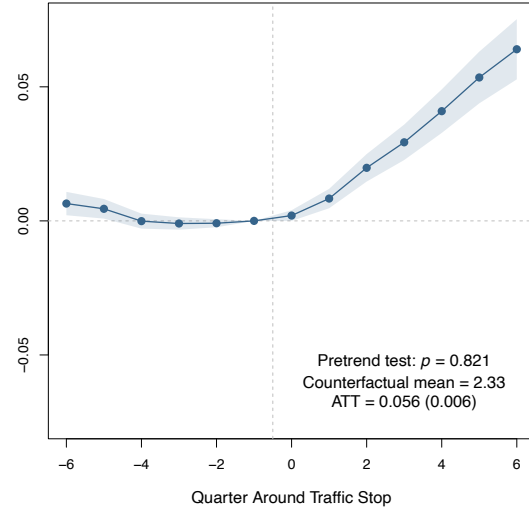
(a) Any New Default



(b) Collections Balances



(c) Credit Line Delinquencies



Notes: Each figure reports event study estimates, obtained via the [Callaway & Sant'Anna \(2021\)](#) approach, as well as 95 percent confidence intervals based on design-based standard errors from [Roth & Sant'Anna \(2022\)](#), for the denoted outcome. Sample is the full event study sample ($N = 525,646$). Figures also report the p -value from the [Borusyak et al. \(2024\)](#) pretend test, the estimated counterfactual mean at $\tau = 6$, and the static ATT estimate.

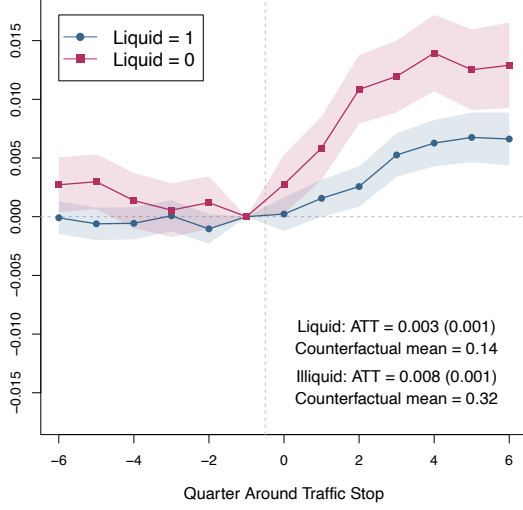
Table 2: Event study estimates for default outcomes

		Collections		Credit Lines	
	(1)	(2)	(3)	(4)	(5)
	Any New Default	Number	Balances	Delinquencies	Derogatories
<i>Event Study Estimates</i>					
$\tau = 1$	0.003 (0.001)	0.01 (0.002)	7.17 (1.84)	0.008 (0.002)	0.005 (0.002)
$\tau = 4$	0.01 (0.001)	0.041 (0.005)	24.64 (3.48)	0.041 (0.004)	0.028 (0.003)
$\tau = 6$	0.009 (0.001)	0.067 (0.006)	35.79 (4.5)	0.064 (0.006)	0.047 (0.005)
ATT	0.005 (0.001)	0.06 (0.006)	33.91 (4.04)	0.056 (0.006)	0.045 (0.004)
<i>Counterfactual Means</i>					
$\tau = 1$	0.22	2.36	1395	2.31	1.57
$\tau = 6$	0.22	2.36	1392	2.32	1.55
<i>Tests for Parallel Trends</i>					
	$p = 0.835$	$p = 0.525$	$p = 0.357$	$p = 0.821$	$p = 0.176$

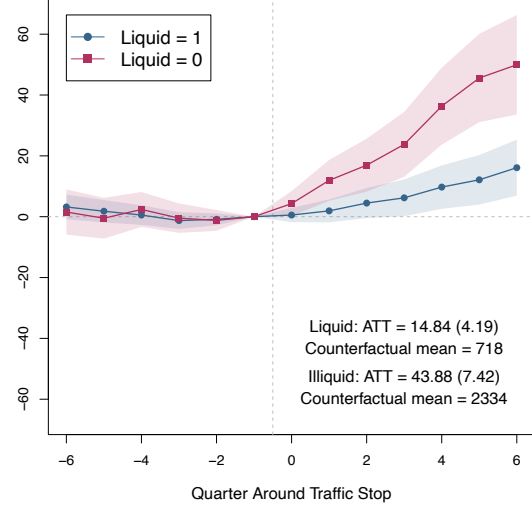
Notes: This table reports event study estimates for one, four, and six quarters post traffic stop, as well as the static ATT estimate, all obtained via the [Callaway & Sant’Anna \(2021\)](#) approach. Design-based standard errors from [Roth & Sant’Anna \(2022\)](#) in parentheses. The lower panels report counterfactual means for $\tau = 1$ and $\tau = 6$, estimated using the method described in the text, and results of the pretrends test from [Borusyak et al. \(2024\)](#). The sample is the full event study sample ($N = 525,646$).

Figure 3: Event study estimates by baseline credit access

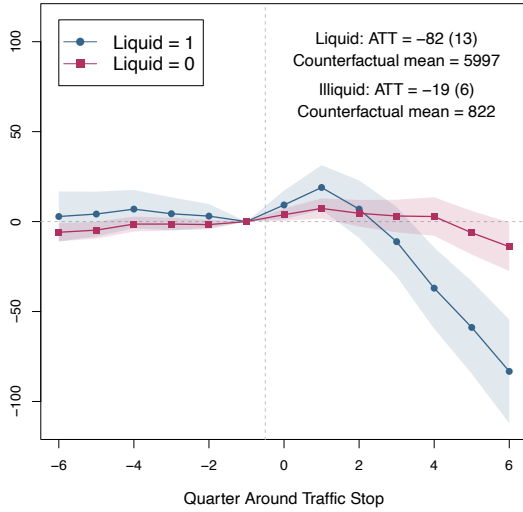
(a) Any New Default



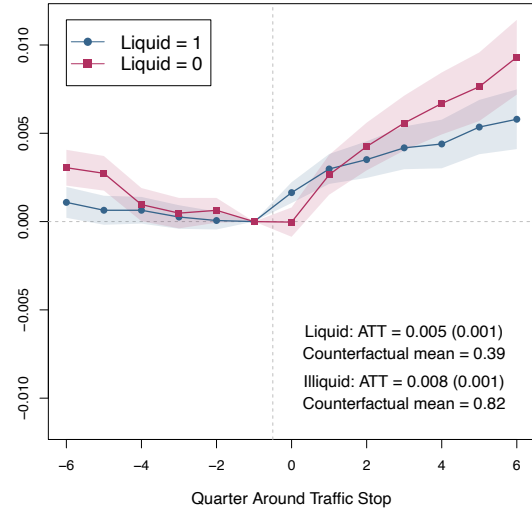
(b) Collections Balances



(c) Revolving Balances

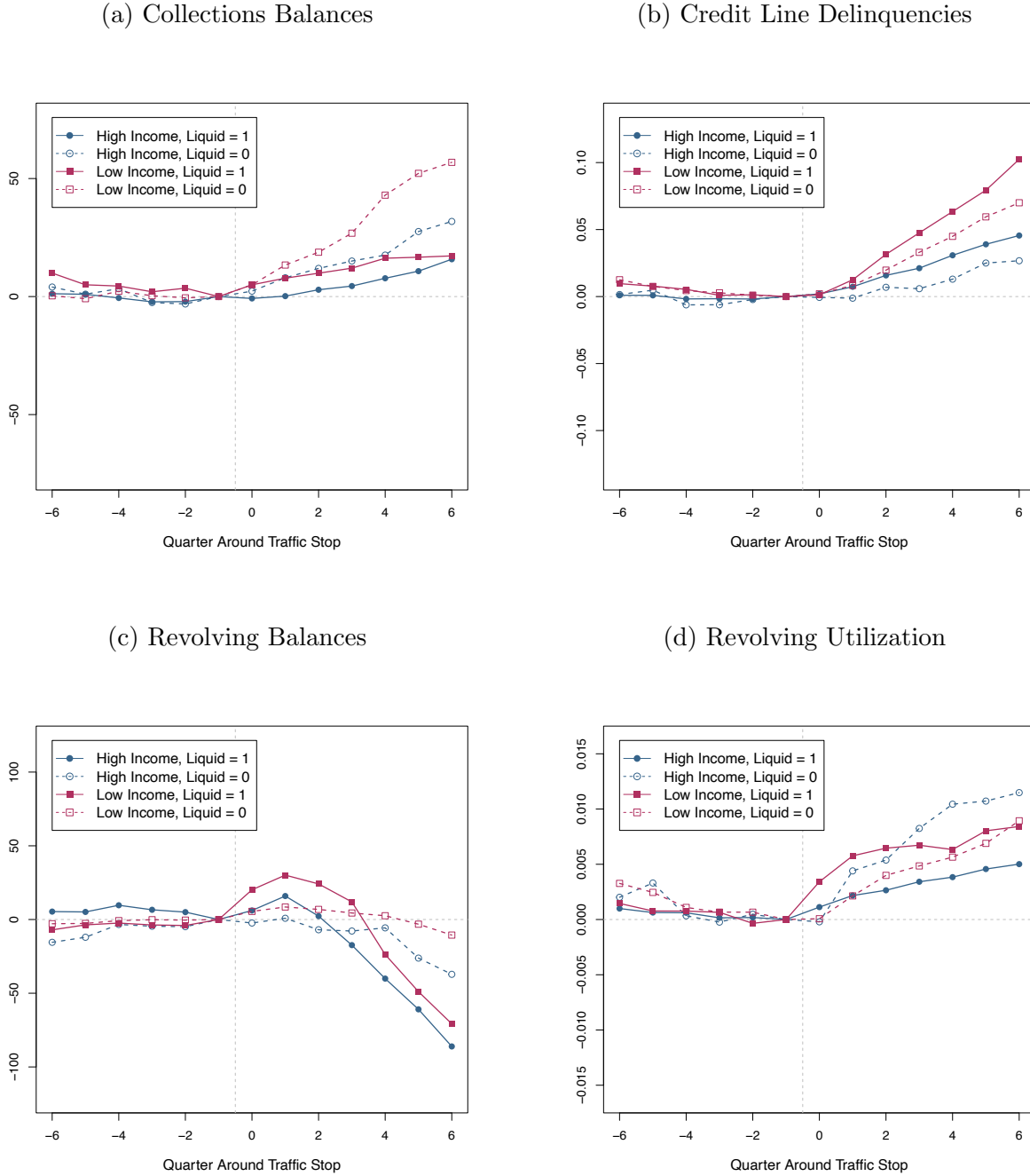


(d) Revolving Utilization



Notes: Each figure reports event study estimates, obtained via the [Callaway & Sant'Anna \(2021\)](#) approach, as well as 95 percent confidence intervals based on design-based standard errors from [Roth & Sant'Anna \(2022\)](#), for the denoted outcome. Event studies are estimated separately for subgroups of motorists based on baseline credit card situation. *Liquid* = 1 is the subset of individuals with at least \$200 in available credit card borrowing at baseline ($N = 301,318$) and *Liquid* = 0 is the subset of individuals with less than \$200 available at baseline, which includes those with no open credit cards at baseline ($N = 224,328$).

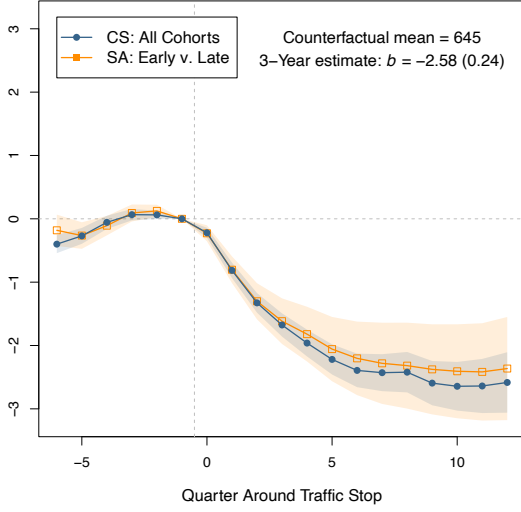
Figure 4: Event study estimates by baseline credit access and estimated income



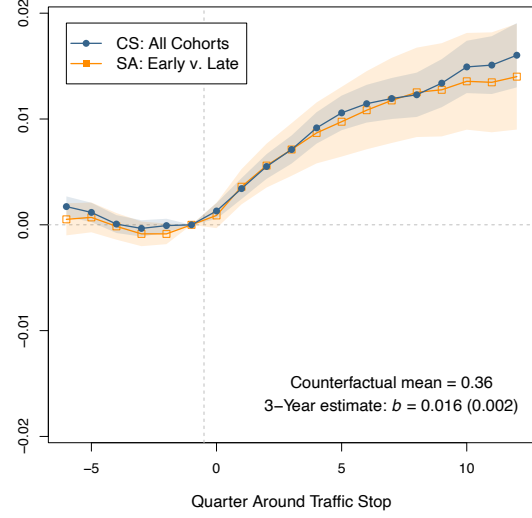
Notes: Each figure reports event study estimates, obtained via the [Callaway & Sant'Anna \(2021\)](#) approach for the denoted outcome. Event studies are estimated separately for subgroups of motorists based on baseline credit card situation and estimated income. *High income* is defined as being above the median baseline estimated income and *liquid* is defined as in figure 3. Sample sizes are $N = 232,230$ (high income, liquid=1), $N = 56,046$ (high income, liquid=0), $N = 69,088$ (low income, liquid=1), $N = 224,328$ (low income, liquid=0).

Figure 5: Event study estimates for long-run outcomes

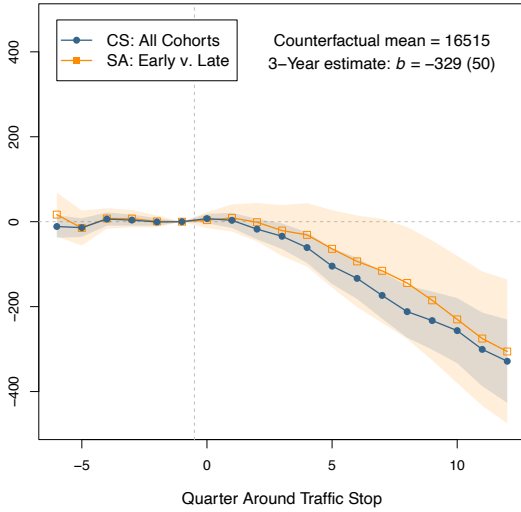
(a) Credit Score



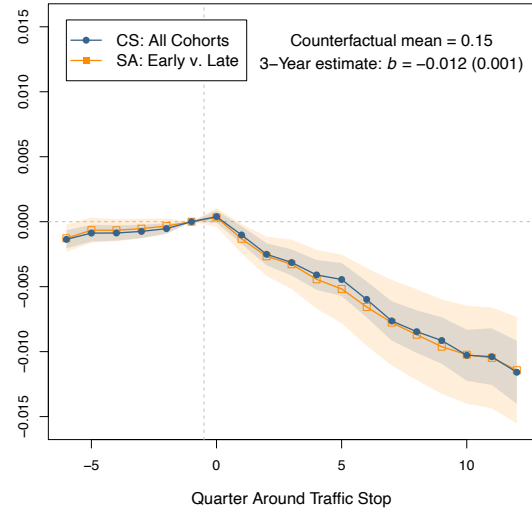
(b) Subprime



(c) Borrowing Limit



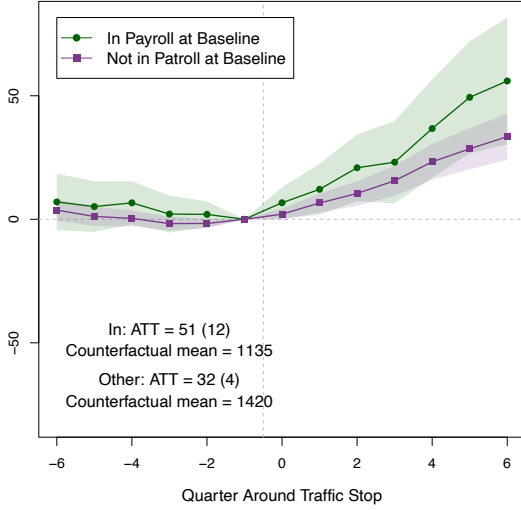
(d) Payroll Employment



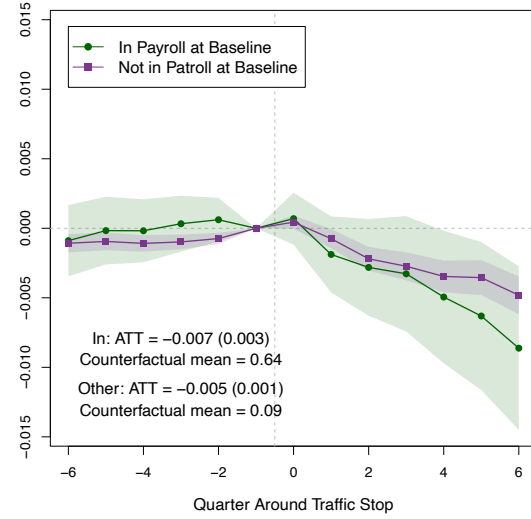
Notes: This figure reports event study estimates obtained from the primary [Callaway & Sant'Anna \(2021\)](#) specification (blue circles) as well as from a specification based on [Sun & Abraham \(2021\)](#) which compares only those cited in 2011–2012 (“early” cohorts) to those cited in 2015Q4 (“late” cohort). Subprime = $\mathbf{1}[\text{credit score} < 600]$. Borrowing limit is the sum of the observed limits across all revolving accounts and thus equals zero for individuals with no revolving lines.

Figure 6: Event study estimates by baseline payroll status

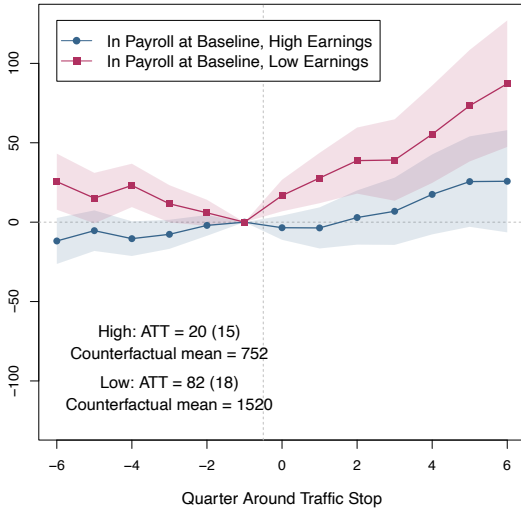
(a) Collections Balances



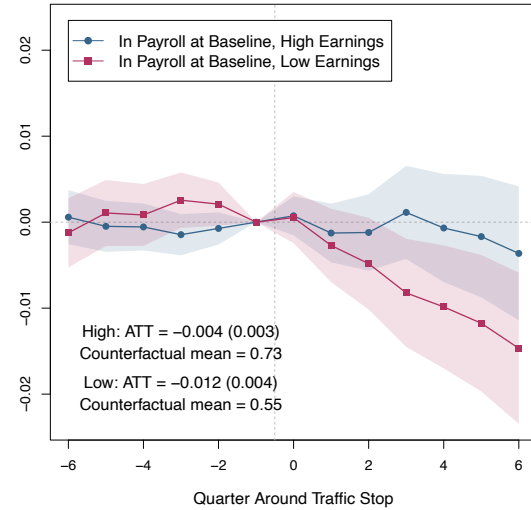
(b) Payroll Employment



(c) Collections Balances

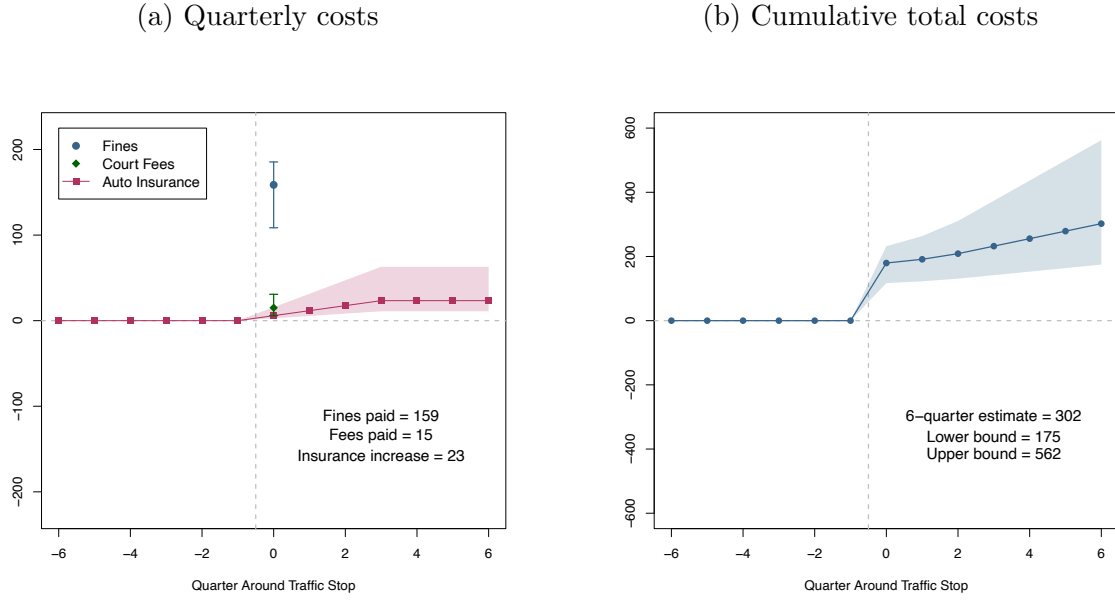


(d) Payroll Employment



Notes: Each figure reports event study estimates, obtained via the [Callaway & Sant'Anna \(2021\)](#) approach, as well as 95 percent confidence intervals based on design-based standard errors from [Roth & Sant'Anna \(2022\)](#), for collections balances or payroll employment, estimated separately by baseline payroll employment status. In panels (a) and (b), the sample is split by whether a motorist was “employed” at baseline, defined as being in the payroll records for all four quarters of 2010 (N employed = 55,140, N other = 470,506). In panels (c) and (d), the sample is the baseline “employed” sample and is split at the median of earnings in 2010, which is \$34,198 (N low = 27,570; N high = 27,570).

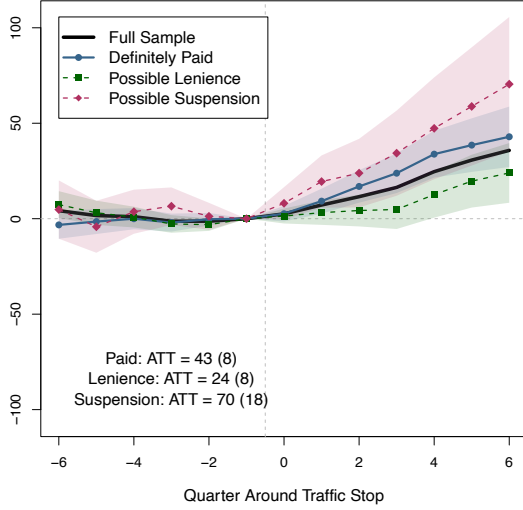
Figure 7: Estimated total costs of citations



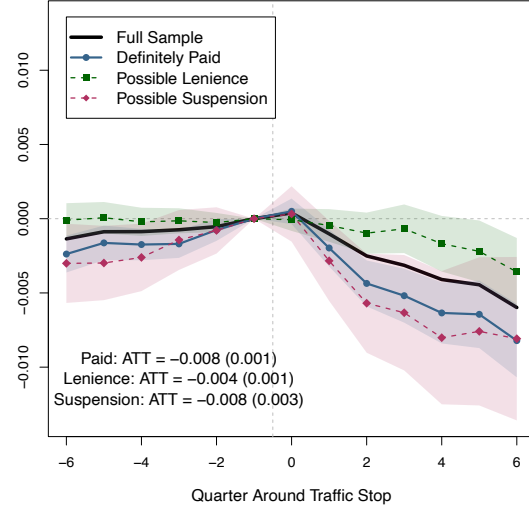
Notes: This figure reports estimated average total costs of citations in the event study sample ($N = 525,646$), taking into account statutory sanctions and the post-citation choices of motorists based on the traffic court dispositions data. Panel (a) reports the per-quarter cost estimates, with estimated fine payments shown in blue circles, estimated court fees paid shown in green diamonds, and car insurance *increases* shown in red squares. Panel (b) reports cumulative total cost estimates. Confidence bands reflect the sensitivity of the estimates to various assumptions. For additional details on the underlying calculations, see appendix C

Figure 8: Event study estimates by traffic court disposition

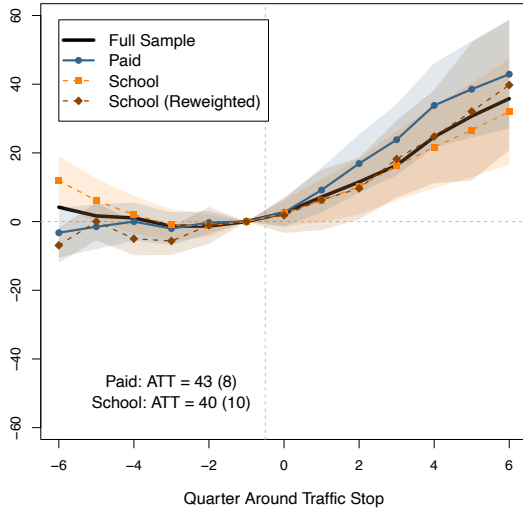
(a) Collections Balances



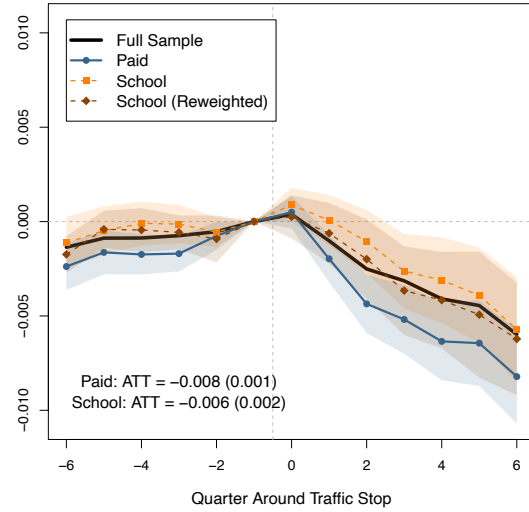
(b) Payroll Employment



(c) Collections Balances



(d) Payroll Employment



Notes: Each figure reports event study estimates, obtained via the [Callaway & Sant'Anna \(2021\)](#) approach, as well as 95 percent confidence intervals based on design-based standard errors from [Roth & Sant'Anna \(2022\)](#), for collections balances or payroll employment, estimated separately by traffic court disposition. In all panels, solid black line in the estimate for the full sample ($N = 525,646$) and solid blue line with circles is the subgroup whose dispositions indicate fine payment ($N = 174,766$; note that this excludes the traffic school group and is therefore a subset of the “definitely paid” sample). In panels (a) and (b), green squares report estimates for those who *may* have received punishment reductions ($N = 175,051$) and red diamonds report estimates for those who *may* have received a license suspension for nonpayment ($N = 40,997$). In panels (c) and (d), orange squares report estimates for those who elected traffic school ($N = 134,832$) and brown diamonds report estimates for the same subsample after reweighting to match the distribution of baseline characteristics (age, gender, race, and quartile bins of credit score and estimated income) in the benchmark payer subsample. See main text and appendix C for additional details.

Table 3: Reweighted ATT estimates and implied affected share estimates

	(1) Baseline	Reweighted to match:				
		Demographics		+ Zip Income		Credit Score
		(2) FL	(3) US	(4) FL	(5) US	(6) US
ATT Estimate	33.91 (4.04)	27.74 (3.51)	29.09 (2.95)	27.7 (4.57)	30.1 (4.31)	28.39 (3.52)
Lower Bound Affected Share (Using estimated $\bar{\Delta} = 315$)	0.108 (0.03)	0.088 (0.025)	0.092 (0.025)	0.088 (0.026)	0.096 (0.028)	0.09 (0.025)
Lower Bound Affected Share (Using estimated $\bar{\Delta} = 112$)	0.302 (0.059)	0.247 (0.049)	0.259 (0.048)	0.247 (0.056)	0.268 (0.056)	0.253 (0.05)
Lower Bound Affected Share (Using estimated $\bar{\Delta} = 89$)	0.383 (0.109)	0.313 (0.09)	0.329 (0.091)	0.313 (0.096)	0.34 (0.1)	0.321 (0.092)

Notes: This table reports ATT estimates for collections balances and associated lower bound affected share estimates (referred to as $\underline{\pi}$ in section 6.3) for the event study sample. Column 1 reports the baseline estimates, while columns 2-6 report estimates when reweighting the sample to match the distribution of baseline characteristics in Florida and the United States. Columns 2-3 reweight the sample to match the age $\times$ gender $\times$ race sample in the 2010 ACS microdata. Columns 3-4 reweight the sample to match the age, race, gender and zip code income distribution by estimating the ATT separately for age $\times$ gender $\times$ race groups, reweighting to match the distribution of zip code income in the Florida and the U.S. for that group, and then aggregating up these estimates, weighting by each group's sample share in Florida and the U.S. Column 6 reports estimates when reweighting only to match the U.S. credit score distribution. Rows 2-4 report the estimated lower bound on the affected share, $\underline{\pi}$, defined as the $ATT/\bar{\Delta}$, using different estimates of the treatment effect upper bound $\bar{\Delta}$ as described in the text and in appendix D. Standard errors for the $\underline{\pi}$ estimates are computed via the delta method.